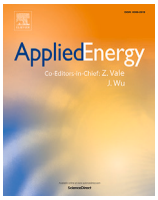
Applied Energy 426 (2026) 128602



Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apen



Bi-timescale distributionally robust optimization for power-computing-carbon coordination in multi-microgrids with data centers

Chenxian Song [a], Hui Chen [a], Jian Wang [b], Yichen Sun [a], Chunyang Gong [a,*], Dikshit Chauhan [c], Chuanlin Zhang [a], Dipti Srinivasan [c]

[a] Intelligent Autonomous Systems Lab, Shanghai University of Electric Power, Shanghai, 200090, China
[b] Smart Energy Innovation College, Shanghai Jiao Tong University, Shanghai, 200240, China
[c] Department of Electrical and Computer Engineering, National University of Singapore, Singapore, 119260, Singapore

HIGHLIGHTS

- Soft actor-critic determines day-ahead server counts.
- Workload migration and batch deferral reshape data-center power demand.
- Wasserstein distributionally robust scheduling uses constraint screening.
- Carbon emission rights and tradable green certificates enter coordinated dispatch.

ARTICLE INFO

Keywords:
Carbon trading
Data center
Multi-microgrid
Spatiotemporal flexibility
Wasserstein distributionally robust chance constraint

ABSTRACT

Rising electricity demand from data centers is increasing economic, reliability and carbon pressure on multi-microgrid operation, while renewable intermittency and workload randomness introduce coupled supply–demand uncertainty. Coordinated scheduling is therefore needed to better exploit data center workload flexibility for risk-controlled and low-carbon multi-microgrid operation. Based on spatial workload migration, temporal batch deferral and cross-microgrid energy sharing, a bi-timescale distributionally robust framework is proposed for power-computing-carbon coordination. At the day-ahead timescale, soft actor-critic (SAC) determines integer server counts, which are fixed in quasi-real-time scheduling to preserve convexity. At the quasi-real-time timescale, data center workload scheduling, microgrid dispatch, service-level agreement (SLA) requirements, carbon emission rights and tradable green certificates (TGCs) are jointly optimized. Uncertainties in renewable generation, local load and workload arrivals are modeled using a Wasserstein distributionally robust chance-constrained (Wasserstein-DRCC) formulation with constraint screening. The decomposed problem is solved by proximal adaptive alternating direction method of multipliers (ADMM), and coalition savings are allocated through sampled Shapley settlement based on risk-aware dual prices. Case studies on five microgrids and five data centers show reductions in daily operating cost, carbon emission and SLA penalties of 32.3%, 27.7% and 75.4%, respectively, compared with the baseline case. Wasserstein-DRCC improves out-of-sample reliability, and SAC-based provisioning reduces daily aggregate idle information technology power by about 13%. These results show that data center workload flexibility can serve as a schedulable resource for economical, low-carbon and reliable multi-microgrid operation.

* Corresponding author.
Email address: gongchunyang@shiep.edu.cn (C. Gong).

https://doi.org/10.1016/j.apenergy.2026.128602
Received 26 April 2026; Received in revised form 11 July 2026; Accepted 31 July 2026
Available online 6 August 2026
0306-2619/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Nomenclature

$\mathcal{M}, m, \mathcal{T}_h, t$ Set/index of MGs; fast-timescale periods/index
$P^{\text{gen}}_{m,t}, P^{\text{ren}}_{m,t}$ Dispatchable generation/renewable utilization power
$P^{\text{dis}}_{m,t}, P^{\text{ch}}_{m,t}$ ESS discharging/charging power
$P^{\text{buy}}_{m,t}, P^{\text{sell}}_{m,t}$ Grid purchase/sale power
$P^{\text{in}}_{m,t}, P^{\text{out}}_{m,t}$ Mutual-support inflow/outflow power
$P^{\text{load}}_{m,t}$ Local non-DC load
$\mathcal{D}_m, d, P^{\text{DC}}_{d,t}$ DCs of MG m; DC index; DC power
$\overline{P}^{\text{gen}}_m, \overline{P}^{\text{ren}}_{m,t}$ Generation/renewable utilization limits
$\overline{P}^{\text{buy}}_m, \overline{P}^{\text{sell}}_m$ Grid purchase/sale limits
$\pi^{\text{sell}}_t, \pi^{\text{buy}}_t$ Grid sale/purchase prices
$\overline{P}^{\text{ex}}_m$ Mutual-support exchange limit
$\overline{P}^{\text{ch}}_m, \overline{P}^{\text{dis}}_m$ ESS charging/discharging limits
$\text{SOC}_{m,t}, \text{SOC}_{m,t-1}$ ESS state of charge
Δt Time-interval length
$\eta^{\text{ch}}_m, \eta^{\text{dis}}_m$ ESS charging/discharging efficiencies
E^{ess}_m ESS energy capacity
$\underline{\text{SOC}}_m, \overline{\text{SOC}}_m$ SOC lower/upper bounds
$\text{SOC}_{m,0}, \text{SOC}^0_m$ Initial SOC variable/value
$\mathcal{W}^{\text{eff}}_{d,t}, \mathcal{W}^{\max}_d$ Effective/maximum delay windows
$|\mathcal{T}_h|, \mathcal{D}$ Horizon length; set of DCs
$A_{d,t}$ Total workload arrivals
$\lambda^{\text{int}}_{d,t}, \lambda^{\text{bat}}_{d,t}$ Nominal interactive/batch workload arrivals used for scheduling
d', τ, τ' DC and workload-time indices
$\nu_{d\to d',t}, \nu^{m}_{d\to d',t}, \nu^{m'}_{d\to d',t}$ Interactive assignment; source/destination local representations
$w^{\text{drop,int}}_{d,t}, w^{\text{drop,bat}}_{d,t}$ Unserved interactive/batch workload
$q_{d\to d',t\to\tau}, q^{m}_{d\to d',t\to\tau}$ Batch assignment; source local representations
$q^{m'}_{d\to d',t\to\tau}, W^{\text{proc}}_{d,t}$ Destination local representations; workload processed by DC d
$\mu_d, \bar{S}_{d,h}, h$ Server service rate; fixed server-count parameter; slow-time index
$b^{\text{int}}_{\text{req}}, b^{\text{bat}}_{\text{req}}, B^{\max}_{d\to d',t}$ Data volume per interactive/batch request; reserved WAN capacity
$P^{\text{IT}}_{d,t}, p^{\text{idle}}_d$ IT power; idle power per active server
κ_d, PUE_d Workload-power coefficient; PUE
$P^{\text{DC},0}_{d,t}, p^{\text{DC},+}_{d,t}, p^{\text{DC},-}_{d,t}$ Baseline DC power; nonnegative power-reduction/increase components
$e_{m,t}$ Carbon emission of MG m in period t
$\xi^{\text{grid}}_t, \xi^{\text{gen}}_m$ Grid/local-generation emission factors
$B^{\text{CER}}_{m,t}, B^{\text{TGC}}_{m,t}$ Purchased CER/TGC quantities
E^{cap}_m Allocated carbon-emission quota
γ^{TGC} Carbon-equivalent conversion coefficient of TGC
$C^{\text{gen}}_m(P^{\text{gen}}_{m,t})$ Local generation operation cost
a_m, b_m, c_m Generation cost coefficients
$C^{\text{ess}}_m(P^{\text{ch}}_{m,t}, P^{\text{dis}}_{m,t})$ ESS operation cost
c^{deg}_m ESS degradation cost coefficient
C^{stf} DC spatiotemporal-flexibility cost
$\pi^{\text{DC},+}_t, \pi^{\text{DC},-}_t$ DC power-reduction/increase settlement prices
$c^{\text{mig}}_{\text{int}}, c^{\text{mig}}_{\text{bat}}$ Interactive/batch migration costs
C^{SLA} SLA penalty cost
$\pi^{\text{drop}}_{\text{int}}, \pi^{\text{drop}}_{\text{bat}}$ Interactive/batch unserved-workload penalties
C^{carbon} Carbon-related procurement cost
$\pi^{\text{CER}}_t, \pi^{\text{TGC}}_t$ CER/TGC procurement prices
C^{econ} Economic operating cost
Ξ, J Scheduling decision variables; scalarized objective
$w^{\text{carbon}}, w^{\text{sla}}$ Carbon/SLA scalarization weights
$\boldsymbol{\xi}_t, n_{\xi}$ Joint uncertainty vector; its dimension
$\tilde{P}^{\text{load}}_{m,t}, \tilde{P}^{\text{ren}}_{m,t}$ Uncertain non-DC load/renewable generation
$\tilde{\lambda}^{\text{int}}_{d,t}, \tilde{\lambda}^{\text{bat}}_{d,t}$ Uncertain interactive/batch arrivals
$\hat{\mathbb{P}}_t$ Empirical distribution
$N_t, n, \hat{\boldsymbol{\xi}}^{(n)}_t$ Sample count/index; uncertainty sample
$\delta_{\hat{\boldsymbol{\xi}}^{(n)}_t}$ Dirac measure at $\hat{\boldsymbol{\xi}}^{(n)}_t$
$\mathcal{P}_t(\rho_t), \rho_t$ Wasserstein ambiguity set; radius
$\mathbb{P}, \mathcal{P}(\mathcal{X}_t), \mathcal{X}_t$ Distribution; distribution set/support
$W_1(\mathbb{P}, \hat{\mathbb{P}}_t)$ First-order Wasserstein distance
$a_{i,t}(\Xi), \mathbf{b}_{i,t}$ Deterministic term; uncertainty coefficient vector
$g^{\text{bal}}_{m,t}(\Xi, \boldsymbol{\xi}_t)$ Net-supply violation function
$g^{\text{sla}}_{d,t}(\Xi, \boldsymbol{\xi}_t)$ Interactive-SLA violation function
$\mathcal{I}, i$ Risk-controlled constraint set/index
$g_{i,t}(\Xi, \boldsymbol{\xi}_t), \epsilon_{i,t}$ Risk-controlled violation function; violation probability
$\theta_{i,t}, [x]_+$ CVaR threshold; positive-part operator
$s^{(n)}_{i,t}, \|\mathbf{b}_{i,t}\|_\infty$ Sample-excess variable; infinity norm
$a^{\text{bal}}_{m,t}(\Xi)$ Balance deterministic term
$\theta^{\text{bal}}_{m,t}, s^{\text{bal},(n)}_{m,t}$ Balance CVaR threshold/excess variable
$\epsilon^{\text{bal}}_{m,t}, \mathbf{b}^{\text{bal}}_{m,t}$ Balance violation probability/coefficient vector
$\hat{P}^{\text{load},(n)}_{m,t}, \hat{P}^{\text{ren},(n)}_{m,t}$ Load/renewable generation samples
$a^{\text{sla}}_{d,t}(\Xi)$ Interactive-SLA deterministic term
$\theta^{\text{sla}}_{d,t}, s^{\text{sla},(n)}_{d,t}$ SLA CVaR threshold/excess variable
$\epsilon^{\text{sla}}_{d,t}, \mathbf{b}^{\text{sla}}_{d,t}$ SLA violation probability/coefficient vector
$\hat{\lambda}^{\text{int},(n)}_{d,t}$ Interactive-arrival sample
$\zeta_{i,t}(\Xi)$ Constraint-screening indicator
$\delta_{i,t}, \Xi^{(0)}$ Screening safety margin; reference solution
$\mathcal{I}^{\text{eff}}_h$ Effective constraint-time pair set
$S_{d,h}, \mathcal{T}^{\text{slow}}$ Integer server-count decision; slow-timescale periods
$\mathbf{s}_h, n_s$ State vector; its dimension
$\hat{\mu}^{\text{int}}_{d,h}, \hat{\sigma}^{\text{int}}_{d,h}$ Predictive mean/std. of interactive arrivals
$\hat{\mu}^{\text{bat}}_{d,h}, \hat{\sigma}^{\text{bat}}_{d,h}$ Predictive mean/std. of batch arrivals
$\hat{\mu}^{\text{ren}}_{m,h}, \hat{\sigma}^{\text{ren}}_{m,h}$ Predictive mean/std. of renewable generation
$\hat{\mu}^{\text{load}}_{m,h}, \hat{\sigma}^{\text{load}}_{m,h}$ Predictive mean/std. of non-DC load
$\mathbf{a}_h, S^{\min}_d, S^{\max}_d$ Server-count action; lower/upper bounds
$\mathbf{s}_{h+1}, \boldsymbol{\omega}_{h+1}$ Next state; next predictive-statistics vector
$\hat{C}^{\text{op}}_h, \Xi^*$ Operating-cost feedback; fast-timescale solution
Φ^{DRCC}_h Wasserstein-DRCC feasibility status
$\theta^*_{i,t}, s^{(n),*}_{i,t}$ Optimized CVaR threshold/excess variable
$\hat{C}^{\text{SLA}}_h$ SLA-penalty feedback
r_h Instantaneous reward
$\psi^{\text{econ}}, \psi^{\text{infeas}}, \psi^{\text{SLA}}$ Economic/infeasibility/SLA reward weights
$\Phi^{\max}$ Infeasibility penalty constant
ϵ_0 Zero-division avoidance constant
$J(\pi_\theta), \pi_\theta$ Maximum-entropy objective; SAC policy
$|\mathcal{T}^{\text{slow}}|, \gamma, \alpha$ Slow-horizon length; discount factor; entropy temperature
$\mathbf{z}_h$ Continuous latent action
$\boldsymbol{\mu}_\theta(\mathbf{s}_h), \boldsymbol{\sigma}_\theta(\mathbf{s}_h)$ Policy mean/std. outputs
$\boldsymbol{\eta}_h, \mathcal{N}(\mathbf{0}, \mathbf{I}_{|\mathcal{D}|})$ Gaussian noise; standard normal distribution
$z_{d,h}, \sigma(z_{d,h})$ Latent action component; sigmoid mapping
$\partial S_{d,h}/\partial z_{d,h}$ STE surrogate derivative
y_h Temporal-difference target
$j, Q_{\bar{\phi}_j}, \bar{\phi}_j$ Critic index; target critic/parameters
$\mathbf{a}'_{h+1}, \mathbf{z}'_{h+1}$ Next integer/latent actions
$\mathcal{L}(\phi_j), \mathcal{R}$ Critic loss; replay buffer
Q_{ϕ_j}, ϕ_j Critic function/parameters
$\mathcal{L}(\theta)$ Actor loss over policy parameters
$\mathcal{L}(\alpha), \mathcal{H}^{\text{tar}}$ Temperature loss; target entropy
τ_{tar} Target-network update coefficient
$\Xi_m, \Xi^{\text{MG}}_m, \Xi^{\text{DC}}_m, \Xi^{\text{DRCC}}_m$ Local decision-variable set and MG/DC/DRCC subsets
$\mathcal{I}^{\text{eff}}_{m,h}, J_m(\Xi_m)$ Local effective constraint-time pair set; local objective

$C_m^{\text{econ}}, C_m^{\text{carbon}}, C_m^{\text{SLA}}$ Economic/carbon/SLA costs
$u_{m,t}(\Xi_m), \mathbf{u}_m(\Xi_m), \mathbf{u}(\Xi)$ Net support injection; local/stacked boundary vectors
$v_{m,t}, \mathbf{v}_m, \mathbf{v}$ Support component; local/stacked consensus vectors
$\Omega_m, C, \text{Proj}_C$ Local feasible set; consensus set; Euclidean projection
$\mathcal{L}_{\beta^k}, \varpi_{m,t}, \boldsymbol{\varpi}_m, \boldsymbol{\varpi}, \beta^k, k$ Augmented Lagrangian; support dual; dual vectors; penalty/iteration
Ξ_m^{k+1}, Ξ_m^k Local ADMM iterates
$\boldsymbol{\varpi}_m^k, \boldsymbol{\varpi}^k$ Previous local/stacked dual iterates
$v_{m,t}^k$ Previous consensus-copy iterate
$\eta_m, \left\|\Xi_m - \Xi_m^k\right\|_2^2$ Proximal coefficient/term
$\mathbf{v}_m^{k+1}, \mathbf{v}^{k+1}$ Updated local/stacked consensus iterates
$|\mathcal{M}|, m'$ Number of MGs; counterpart MG index
$\boldsymbol{\varpi}_m^{k+1}$ Updated local dual iterate
$r_{\text{pri}}^{k+1}, r_{\text{dual}}^{k+1}$ Primal/dual consensus residuals
$\beta^{k+1}, \tau_{\text{inc}}, \tau_{\text{dec}}, \mu_{\text{ad}}$ Updated penalty; residual-balancing parameters
$\epsilon_{\text{pri}}, \epsilon_{\text{dual}}$ Primal/dual stopping tolerances
Ξ_m^*, J_m^*, J^* Converged local schedule; coordinated local/system costs
$\boldsymbol{\varpi}_m^*, \varpi_{m,t}^*$ Converged dual vector; pricing component
$J_m^{(0)}, J^{(0)}$ Local/total standalone scalarized costs
Π Total coalition cost saving
$S, \mathcal{V}(S), J^*(S)$ Coalition; coalition value; optimized coalition cost
$S, \mathcal{V}(S), J^*(S)$ Coalition; coalition value; optimized coalition cost
$\phi_m, |S|$ Shapley value; coalition size
$\{\sigma^r\}_{r=1}^K, r, K$ Sampled permutations; index/count
$P_m^{\sigma^r}$ Predecessor set of MG m in σ^r
$\hat{\phi}_m$ Sampled Shapley allocation
J_m^{sett} Post-settlement cost of MG m

1. Introduction

1.1. Background and motivation

Global electricity consumption by data centers was estimated at approximately 415 TWh in 2024 and is projected to reach about 945 TWh by 2030 in the International Energy Agency (IEA) Base Case [1]. This growth is being driven by the rapid expansion of artificial-intelligence-related computing infrastructure and digital services. Although the uncertainty and heterogeneity of data-center energy estimates have been systematically documented [2], the upward demand trend is clear. As a result, increasing pressure is being placed on power supply security and carbon-neutrality targets, making data centers critical flexible loads for coordinated power-computing-carbon management. The integration of data centers with multi-microgrid (MMG) systems offers a promising pathway for coordinated low-carbon operation. In networked multi-microgrid systems, renewable-energy utilization, energy sharing, and mutual operational support among distributed energy resources can be enhanced through coordinated energy management and transactive energy mechanisms [3,4]. Moreover, the delay-tolerance heterogeneity of data-center workloads, where interactive services must be processed immediately and batch services can be deferred within allowable service deadlines, provides spatiotemporal flexibility (STF) that reshapes power demand across both spatial and temporal dimensions [5,6]. Such spatiotemporal flexibility constitutes a demand-side resource that is structurally distinct from conventional flexible loads.

Realizing this potential requires three coupled challenges to be addressed. First, server capacity is configured as an integer day-ahead decision, whereas workload scheduling in data centers is performed at a quasi-real-time timescale; their joint optimization can compromise the convex structure required for tractable distributionally robust reformulations [7]. Second, uncertainty from renewable generation, local loads, and heterogeneous workload arrivals cannot be fully characterized by deterministic optimization or scenario-based stochastic programming when out-of-sample reliability is required [8,9]. Third, under China's dual-carbon policy, carbon emission rights (CERs) and tradable green certificates (TGCs) must be embedded into operational scheduling, whereas their interactions with workload migration and benefit allocation in multi-microgrid coalitions remain insufficiently characterized [10,11]. These challenges motivate the bi-timescale power-computing-carbon coordination framework developed in this paper.

1.2. Literature review and research gap

Distributed energy management in multi-microgrid systems has been widely studied using the alternating direction method of multipliers (ADMM), by which a global optimization problem is decomposed into privacy-preserving local subproblems coupled through consensus variables [12]. Robust and adaptive ADMM variants have also been developed for decentralized microgrid coordination [13], while Shapley-value-based cooperative game methods have been used for benefit allocation in energy communities and multi-microgrid scheduling [14,15]. However, data centers have generally been treated as fixed loads in these studies. The joint spatial migration and temporal deferral of data-center workloads, together with their implications for distributionally robust multi-microgrid coordination and spatiotemporal flexibility settlement, remain insufficiently characterized.

On the side of data centers, shared energy-storage business models for clusters of data centers under renewable uncertainty have been investigated, as have robust multi-objective operational models for islanded microgrids with data centers [16,17]. The demand-side flexibility of cloud data centers has also been incorporated into bi-level planning under electricity-carbon markets [18]. However, the joint use of spatial workload migration and temporal workload deferral remains insufficiently characterized in distributionally robust coordination for multi-microgrid systems with data centers. More importantly, when integer server capacity is retained in the quasi-real-time scheduling layer, discrete server counts are coupled with continuous workload scheduling variables in the power model of data centers. The resulting Wasserstein distributionally robust chance-constrained (Wasserstein-DRCC) scheduling problem becomes a mixed-integer program, increasing computational complexity relative to existing distributionally robust power-system and networked microgrid formulations [19,20]. Meanwhile, deep reinforcement learning has been applied to dispatch of microgrids, and the soft actor-critic (SAC) algorithm has been used in sequential energy management [21,22]. However, the use of SAC for day-ahead discrete server capacity preconfiguration in a bi-timescale multi-microgrid framework has received limited attention. Its coupling with Wasserstein-DRCC feasibility feedback from quasi-real-time scheduling also remains underexplored. This methodological gap motivates the bi-timescale decoupling strategy proposed in this paper.

With regard to uncertainty management, Wasserstein-DRCC optimization has been used to enhance data-driven reliability without requiring a prescribed probability distribution. Tractable reformulations can be obtained when the underlying scheduling model preserves affine or convex structures [23,24]. Related applications have been reported in virtual power plant scheduling [25], cascade hydroplant dispatch [26], and multi-energy microgrid energy management [27]. However, such tractability becomes more difficult to preserve when integer server-capacity decisions are retained in the quasi-real-time scheduling layer of data centers. In this case, discrete server counts are coupled with continuous workload scheduling variables, and the resulting reformulation becomes a mixed-integer program. This structure is less compatible with convex distributed coordination and duality-based coalition pricing. Moreover, sample-indexed auxiliary variables

introduced by Wasserstein-DRCC reformulations can increase with both the number of historical samples and the number of constraints, creating an online computational burden in large-scale multi-microgrid systems. Therefore, scalable constraint screening for Wasserstein-DRCC coordination with data centers remains insufficiently characterized in low-carbon scheduling studies [28].

With respect to low-carbon operation, market-based instruments, including carbon emission rights and TGCs, have been introduced into networked microgrids and multi-microgrid systems to support energy-carbon coordination [29,30]. Coupled carbon-certificate mechanisms and low-carbon bargaining strategies have also been examined in virtual power plant and multi-microgrid operation, where cost and emission reductions are reported under the corresponding case settings [31,32]. However, these studies mainly focus on energy-side trading or coalition operation, whereas the coupling between low-carbon incentive signals and workload migration of data centers has received limited attention. In parallel, spatiotemporal power-computing response and flexibility resources of data centers have been modeled or evaluated [33,34]. Their integration with carbon-accounting decisions, however, remains insufficiently characterized. Allocation methods based on Aumann–Shapley principles and virtual energy-storage sharing provide useful precedents for cost or capacity allocation in power networks [35,36]. Their extension to bi-timescale multi-microgrid coordination remains insufficiently explored, particularly when endogenous risk-aware clearing prices derived from the ADMM and spatiotemporal flexibility settlement costs are considered together.

Based on the foregoing review, four research gaps can be identified. The first concerns the timescale mismatch between integer server-capacity configuration and quasi-real-time Wasserstein-DRCC scheduling. A tractable convex reformulation becomes difficult to retain when these two decision layers are optimized jointly. The second concerns the limited characterization of a unified spatiotemporal flexibility tensor that jointly represents spatial workload migration and temporal deferral within distributed multi-microgrid systems. The third relates to the insufficient integration of carbon emission rights and TGCs with spatiotemporal computing flexibility under privacy-aware distributed coordination. The fourth concerns the limited extension of Shapley-based revenue allocation to endogenous risk-aware clearing prices and spatiotemporal flexibility settlement costs within a bi-timescale architecture. These gaps are made more explicit in the operational setting considered in this work. Geographically distributed data centers are coordinated across regions, interactive and batch services with different delay tolerances are hosted, and dual carbon-market mechanisms are considered. In this setting, server capacity is provisioned ahead of time as an integer decision, whereas workload scheduling is performed at a quasi-real-time timescale. If both decisions are handled within a single timescale, multisource fluctuations in renewable generation, local loads, and workload arrivals may not be adequately absorbed with reliability guarantees. As a result, the power supply assurance of data centers can be affected. This operational difficulty motivates the bi-timescale design developed in the following sections. These gaps are addressed in this paper through a bi-timescale framework that integrates SAC-based server provisioning, Wasserstein-DRCC with constraint screening, carbon emission rights and TGCs-coupled distributed dispatch, and a Shapley allocation mechanism. The specific contributions are detailed in Section 1.3. These components should not be regarded as independently stackable. The day-ahead server count determined by SAC-based provisioning is fixed before quasi-real-time scheduling, which helps preserve the convexity and tractability of the fast-timescale Wasserstein-DRCC problem. The constraint screening mechanism is designed to improve computational scalability in multi-microgrid systems. The converged mutual-support dual components provide risk-aware price signals for coordinated power exchange, while the resulting operating costs provide the basis for Shapley-based allocation. This chain of structural dependencies forms the integration novelty of the proposed framework.

1.3. Contributions and paper organization

To address the four research gaps identified in Section 1.2, a hierarchical power-computing-carbon coordination framework is proposed for data center-integrated multi-microgrid systems. The main contributions are summarized as follows.

1. A bi-timescale decoupling strategy is developed to coordinate server provisioning and robust scheduling. SAC determines day-ahead server counts, which are fixed as quasi-real-time capacity parameters to remove integer–continuous coupling and preserve convexity for Wasserstein-DRCC reformulation and distributed pricing.
2. A spatiotemporal flexibility scheduling model is established to capture data center workload migration and batch deferral in multi-microgrid operation. The model couples workload flexibility with microgrid dispatch, carbon emission rights/TGC mechanisms, and service-level-agreement (SLA) constraints for economic and low-carbon scheduling.
3. A Wasserstein-DRCC model with constraint screening is formulated to manage multisource uncertainty in renewable generation, local load, and workload arrivals. The model controls power-balance and interactive-SLA risks, while the screening strategy improves the tractability of robust scheduling at the multi-microgrid scale.
4. A privacy-preserving distributed coordination and settlement mechanism is developed by combining proximal adaptive ADMM with sampled Shapley allocation. Local microgrid subproblems are coordinated through limited boundary-information exchange, while the mutual-support components of the ADMM dual vectors provide risk-aware price signals for coordinated power exchange.

The remainder of this paper is organized as follows. Section 2 presents the system architecture and coordination mechanisms. Section 3 formulates the power-computing-carbon coordinated optimization model. Section 4 develops the uncertainty modeling and Wasserstein-DRCC reformulation. Section 5 presents the SAC-based day-ahead server capacity provisioning method. Section 6 introduces the distributed coordination and coalition settlement framework, including the proximal adaptive ADMM solution and the Shapley-based settlement of coalition cost savings. Section 7 provides the case studies. Section 8 concludes the paper.

2. System architecture and coordination mechanisms

This section describes the physical architecture and information-exchange structure of the proposed framework. The presentation follows four layers: the coordinated multi-microgrid entities and bi-timescale hierarchy, the data center power-computing coupling that provides spatiotemporal flexibility, the carbon entitlement and certificate coupling that links low-carbon incentives with operational decisions, and the distributed coordination and settlement interface for multi-agent implementation. These four layers are described in Sections 2.1–2.4, respectively. The overall architecture is shown in Fig. 1.

2.1. System entities and bi-timescale coordination architecture

The considered system is a coordinated multi-microgrid system with geographically distributed microgrids and data centers electrically supplied by these microgrids. Cross-microgrid coordination is performed by the multi-microgrid coordination operator, while local operational information, cost functions, and device constraints are retained by each microgrid.

To address the mismatch between the day-ahead server-capacity decision and quasi-real-time workload scheduling, a bi-timescale architecture is adopted. At the slow timescale, the soft actor-critic (SAC) algorithm is used to determine the number of activated servers

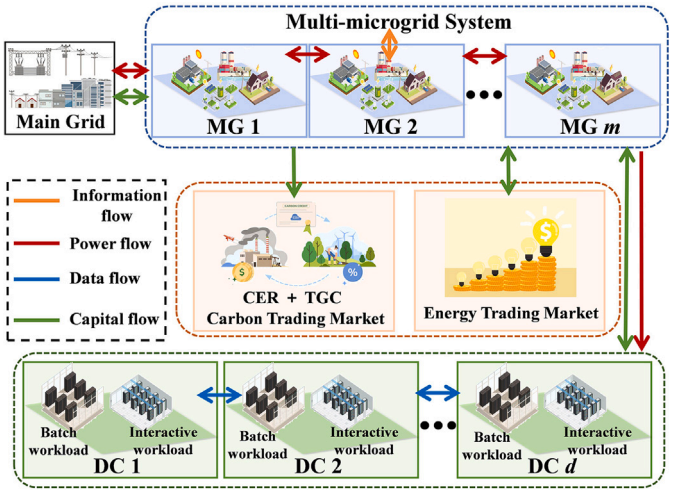


Fig. 1. System architecture for power-computing-carbon coordination in data-center-integrated multi-microgrids.

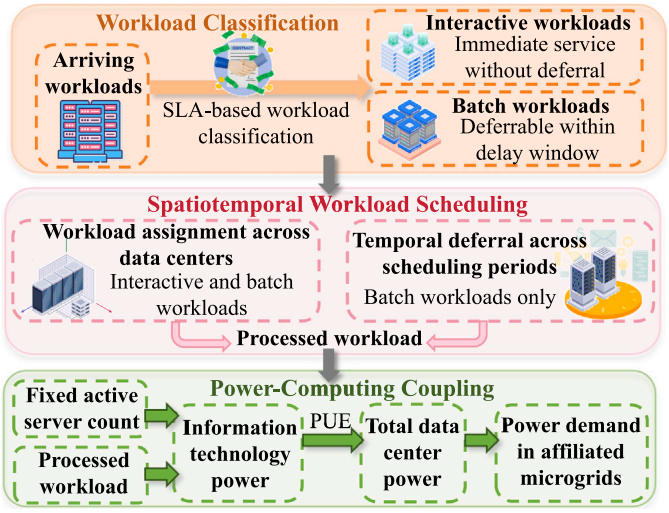


Fig. 2. Data center spatiotemporal workload flexibility and power-computing coupling model.

for each data center. The resulting server count is then passed to the fast timescale as a fixed capacity parameter. At the fast timescale, the Wasserstein distributionally robust chance-constrained (Wasserstein-DRCC) scheduling problem is solved under the fixed server capacity, and the resulting operating cost, feasibility status, and SLA penalty are returned to the SAC agent as reward feedback. This structure prevents the integer server-count decision from being embedded directly in the fast-timescale continuous scheduling problem, thereby preserving the convex fast-timescale structure required by the Wasserstein-DRCC reformulation and distributed solution.

2.2. Data center power-computing coupling and spatiotemporal flexibility

A linear idle-plus-load-proportional server power model, together with power usage effectiveness (PUE), is used to map workload processing and server operating states to the electricity consumption of data centers. Under this representation, data centers are modeled as partially controllable demand-side resources whose electricity demand can be reshaped only within the limits imposed by computing-service requirements. According to the delay tolerance specified by the service level agreement (SLA), workloads are divided into interactive workloads that require immediate service without deferral and batch workloads that can be deferred within a prescribed maximum delay window. This classification provides the operational basis for spatiotemporal flexibility.

The spatiotemporal flexibility mechanism jointly uses the temporal elasticity of batch workloads and the spatial migration capability enabled by the wide-area network (WAN). Through this mechanism, data center power demand can be redistributed across both regions and periods, rather than being adjusted only through spatial migration. Therefore, workload scheduling is linked to renewable accommodation, electricity prices, carbon-related incentives, and service reliability.

The two flexibility dimensions are distinct but coupled. Spatial migration reallocates workload among data centers and is constrained by inter-data-center communication conditions and migration costs. Temporal deferral shifts batch workload to later eligible periods within its delay budget and is not applicable to interactive workloads with strict latency requirements. Consequently, feasible spatiotemporal scheduling depends on both network-aware workload migration and accurate workload delay-tolerance classification.

2.3. Carbon entitlement and certificate coupling

Carbon emission rights and TGCs are incorporated into the coordination framework to represent carbon-related operating costs and low-carbon incentives. The procurement costs and entitlement balances of carbon emission rights and TGCs are modeled so that carbon-related signals can affect generation scheduling and workload migration decisions. This coupling allows low-carbon operational incentives to be considered together with economic dispatch and SLA-oriented service constraints.

2.4. Distributed coordination and settlement interface

In the multi-agent coordination setting, local cost functions, device constraints, workload parameters, and SLA-related tolerance limits are treated as private information. Each microgrid solves its local subproblem independently, while only boundary coordination quantities and dual information are exchanged with the multi-microgrid coordination operator. This information-exchange structure provides the interface for the proximal adaptive alternating direction method of multipliers (proximal adaptive ADMM) developed later.

After distributed scheduling is completed, the economic benefit generated by multi-microgrid cooperation is allocated through a Shapley-value-based settlement mechanism. A permutation-sampling approximation is adopted to avoid exhaustive enumeration of all joining orders. The settlement stage is linked to the distributed coordination result without requiring the disclosure of local private models.

3. Coordinated optimization model for power-computing-carbon coupling

An optimization model is formulated to characterize the coordinated operation of the multi-microgrid system with data centers. As illustrated in Fig. 2, the data center-side coupling model links SLA-based workload classification, spatial migration, temporal deferral, and the conversion from processed workload and server operating state to information technology power and total data center power. On this basis, the formulation incorporates microgrid operating constraints, spatiotemporal flexibility constraints, carbon-related cost accounting, and the scalarized operating objective.

3.1. Microgrid operating constraints

Each microgrid $m \in \mathcal{M}$ may contain dispatchable generation, renewable generation, an energy storage system, local non-data-center load,

and data center load. For $m \in \mathcal{M}$ and $t \in \mathcal{T}_h$, the power balance is:

$$\begin{aligned} &P^{\mathrm{gen}}_{m,t} + P^{\mathrm{ren}}_{m,t} + P^{\mathrm{dis}}_{m,t} + P^{\mathrm{buy}}_{m,t} + P^{\mathrm{in}}_{m,t} \\ &= P^{\mathrm{load}}_{m,t} + P^{\mathrm{ch}}_{m,t} + P^{\mathrm{sell}}_{m,t} + P^{\mathrm{out}}_{m,t} + \sum_{d \in \mathcal{D}_m} P^{\mathrm{DC}}_{d,t}, \quad \forall m \in \mathcal{M},\ t \in \mathcal{T}_h \end{aligned} \tag{1}$$

The dispatchable generation and renewable utilization constraints are:

$$\begin{cases} 0 \le P^{\mathrm{gen}}_{m,t} \le \overline{P}^{\mathrm{gen}}_{m} \\ 0 \le P^{\mathrm{ren}}_{m,t} \le \overline{P}^{\mathrm{ren}}_{m,t} \end{cases} \quad \forall m \in \mathcal{M},\ t \in \mathcal{T}_h \tag{2}$$

The grid transaction constraints are:

$$\begin{cases} 0 \le P^{\mathrm{buy}}_{m,t} \le \overline{P}^{\mathrm{buy}}_{m} \\ 0 \le P^{\mathrm{sell}}_{m,t} \le \overline{P}^{\mathrm{sell}}_{m} \\ \pi^{\mathrm{sell}}_{t} < \pi^{\mathrm{buy}}_{t} \end{cases} \quad \forall m \in \mathcal{M},\ t \in \mathcal{T}_h \tag{3}$$

The price inequality rules out arbitrage profits from simultaneous grid purchase and sale. The mutual-support exchange constraints are:

$$\begin{cases} 0 \le P^{\mathrm{in}}_{m,t} \le \overline{P}^{\mathrm{ex}}_{m} \\ 0 \le P^{\mathrm{out}}_{m,t} \le \overline{P}^{\mathrm{ex}}_{m} \\ -\overline{P}^{\mathrm{ex}}_{m} \le P^{\mathrm{out}}_{m,t} - P^{\mathrm{in}}_{m,t} \le \overline{P}^{\mathrm{ex}}_{m} \end{cases} \quad \forall m \in \mathcal{M},\ t \in \mathcal{T}_h \tag{4}$$

The energy storage system constraints are:

$$\begin{cases} 0 \le P^{\mathrm{ch}}_{m,t} \le \overline{P}^{\mathrm{ch}}_{m} \\ 0 \le P^{\mathrm{dis}}_{m,t} \le \overline{P}^{\mathrm{dis}}_{m} \\ \mathrm{SOC}_{m,t} = \mathrm{SOC}_{m,t-1} + \Delta t(\eta^{\mathrm{ch}}_{m} P^{\mathrm{ch}}_{m,t} - P^{\mathrm{dis}}_{m,t}/\eta^{\mathrm{dis}}_{m})/E^{\mathrm{ess}}_{m} \\ \underline{\mathrm{SOC}}_{m} \le \mathrm{SOC}_{m,t} \le \overline{\mathrm{SOC}}_{m} \\ \mathrm{SOC}_{m,0} = \mathrm{SOC}^{0}_{m} \\ \forall m \in \mathcal{M},\ t \in \mathcal{T}_h \end{cases} \tag{5}$$

The initial state of charge is specified by SOC^{0}_{m}.

3.2. SLA-constrained data center workload scheduling and power-computing flexibility

For each data center, the arriving workload is classified into interactive and batch workloads according to SLA requirements. Interactive workloads are scheduled within the arrival period, whereas batch workloads can be deferred within their admissible delay windows. Spatiotemporal flexibility is therefore represented by workload assignment across data centers and temporal deferral across scheduling periods. These scheduling decisions determine the processed workload of each data center and thereby couple computing-side operations with microgrid power dispatch.

The effective delay window is defined as:

$$\mathcal{W}^{\mathrm{eff}}_{d,t} = \min\{\mathcal{W}^{\max}_{d},\ |\mathcal{T}_h| - t\}, \quad \forall d \in \mathcal{D},\ t \in \mathcal{T}_h \tag{6}$$

The workload conservation constraints are:

$$\begin{cases} A_{d,t} = \lambda^{\mathrm{int}}_{d,t} + \lambda^{\mathrm{bat}}_{d,t} \\ \lambda^{\mathrm{int}}_{d,t} = \sum_{d' \in \mathcal{D}} \nu_{d \to d',t} + w^{\mathrm{drop,int}}_{d,t} \\ \lambda^{\mathrm{bat}}_{d,t} = \sum_{d' \in \mathcal{D}} \sum_{\tau=t}^{t+\mathcal{W}^{\mathrm{eff}}_{d,t}} q_{d \to d',t \to \tau} + w^{\mathrm{drop,bat}}_{d,t} \\ W^{\mathrm{proc}}_{d,t} = \sum_{d' \in \mathcal{D}} \nu_{d' \to d,t} + \sum_{d' \in \mathcal{D}} \sum_{\tau'=\max\{1,\, t-\mathcal{W}^{\max}_{d'}\}}^{t} q_{d' \to d,\tau' \to t} \\ \forall d \in \mathcal{D},\ t \in \mathcal{T}_h \end{cases} \tag{7}$$

The effective delay window prevents batch workloads from being scheduled beyond the horizon. The variable $\nu_{d \to d',t}$ denotes the portion of the nominal interactive workload originating at data center d in period t that is committed to same-period service at data center d'. The variable $q_{d \to d',t \to \tau}$ denotes the batch workload arriving at data center d in period t and assigned to data center d' for processing in period τ. Local execution is included when $d' = d$, whereas cross-data-center scheduling occurs when $d' \neq d$. The variables $w^{\mathrm{drop,int}}_{d,t}$ and $w^{\mathrm{drop,bat}}_{d,t}$ denote unserved interactive and batch workloads, respectively, and are used for SLA penalty evaluation in the objective function.

The workload non-negativity constraints and the data center processing-capacity constraint are:

$$\begin{cases} \nu_{d \to d',t} \ge 0 \\ q_{d \to d',t \to \tau} \ge 0 \\ w^{\mathrm{drop,int}}_{d,t} \ge 0 \\ w^{\mathrm{drop,bat}}_{d,t} \ge 0 \\ 0 \le W^{\mathrm{proc}}_{d,t} \le 3600\Delta t\, \mu_d \bar{S}_{d,h} \end{cases} \tag{8}$$

The first four inequalities are imposed on all admissible workload-scheduling variables. The processing-capacity constraint holds for all $d \in \mathcal{D}$ and $t \in \mathcal{T}_h$. Since $W^{\mathrm{proc}}_{d,t}$ is measured as a period-wise request count, $3600\Delta t\, \mu_d \bar{S}_{d,h}$ gives the available processing capacity within period t.

For cross-data-center workload scheduling, the WAN capacity constraint is:

$$\begin{aligned} &b^{\mathrm{int}}_{\mathrm{req}} \nu_{d \to d',t} + b^{\mathrm{bat}}_{\mathrm{req}} \sum_{\tau=t}^{t+\mathcal{W}^{\mathrm{eff}}_{d,t}} q_{d \to d',t \to \tau} \le B^{\max}_{d \to d',t}\, 3600\Delta t \\ &\forall d, d' \in \mathcal{D},\ d \neq d',\ t \in \mathcal{T}_h \end{aligned} \tag{9}$$

The left-hand side represents the data volume induced by workloads assigned from data center d to data center d' during period t, and the right-hand side represents the reserved WAN transmission capacity over the same period. This constraint ensures that spatiotemporal workload scheduling remains compatible with the available inter-data-center communication capacity.

The data center power model with PUE is:

$$\begin{cases} P^{\mathrm{IT}}_{d,t} = p^{\mathrm{idle}}_{d} \bar{S}_{d,h} + \kappa_d W^{\mathrm{proc}}_{d,t}/(3600\Delta t) \\ P^{\mathrm{DC}}_{d,t} = \mathrm{PUE}_d\, P^{\mathrm{IT}}_{d,t} \end{cases} \quad \forall d \in \mathcal{D},\ t \in \mathcal{T}_h \tag{10}$$

Therefore, workload scheduling affects data center power demand through the processed workload $W^{\mathrm{proc}}_{d,t}$, which links computing-side scheduling decisions to the microgrid energy-balance constraints. A

no-flexibility baseline power profile is defined by assuming local same-period processing of the arriving workloads:

$$P_{d,t}^{\mathrm{DC},0} = \mathrm{PUE}_d \left(p_d^{\mathrm{idle}} \bar{S}_{d,h} + \kappa_d (\lambda_{d,t}^{\mathrm{int}} + \lambda_{d,t}^{\mathrm{bat}}) / (3600 \Delta t) \right) \tag{11}$$

The actual data center power is related to this baseline by:

$$\begin{aligned} & P_{d,t}^{\mathrm{DC}} = P_{d,t}^{\mathrm{DC},0} - p_{d,t}^{\mathrm{DC},+} + p_{d,t}^{\mathrm{DC},-} \\ & p_{d,t}^{\mathrm{DC},+} \geq 0,\ p_{d,t}^{\mathrm{DC},-} \geq 0, \quad \forall d \in \mathcal{D},\ t \in \mathcal{T}_h \end{aligned} \tag{12}$$

By combining the baseline definition with the data center power model, the net deviation is expressed as:

$$p_{d,t}^{\mathrm{DC},+} - p_{d,t}^{\mathrm{DC},-} = \mathrm{PUE}_d \kappa_d \left(\lambda_{d,t}^{\mathrm{int}} + \lambda_{d,t}^{\mathrm{bat}} - W_{d,t}^{\mathrm{proc}} \right) / (3600 \Delta t) \tag{13}$$

Here, $p_{d,t}^{\mathrm{DC},+}$ and $p_{d,t}^{\mathrm{DC},-}$ are defined as the non-negative reduction and increase components of data center power relative to the baseline, respectively. Their net value is determined by Eq. (13).

3.3. Carbon-emission accounting and entitlement constraints

Carbon emission rights and TGCs are incorporated to internalize the carbon consequences of microgrid dispatch. In the proposed coordination model, data center workload scheduling affects carbon emissions indirectly by reshaping data center power demand and thereby changing the grid-purchase and local-generation decisions of microgrids.

The carbon emission of microgrid m in period t is defined as:

$$e_{m,t} = \left(\xi_t^{\mathrm{grid}} P_{m,t}^{\mathrm{buy}} + \xi_m^{\mathrm{gen}} P_{m,t}^{\mathrm{gen}} \right) \Delta t, \quad \forall m \in \mathcal{M},\ t \in \mathcal{T}_h \tag{14}$$

The purchased quantities of carbon emission rights and TGCs are non-negative:

$$B_{m,t}^{\mathrm{CER}} \geq 0, \quad B_{m,t}^{\mathrm{TGC}} \geq 0, \quad \forall m \in \mathcal{M},\ t \in \mathcal{T}_h \tag{15}$$

The accumulated carbon emission of each microgrid is constrained by its allocated quota and purchased carbon-equivalent entitlements:

$$\sum_{t \in \mathcal{T}_h} e_{m,t} \leq E_m^{\mathrm{cap}} + \sum_{t \in \mathcal{T}_h} B_{m,t}^{\mathrm{CER}} + \gamma^{\mathrm{TGC}} \sum_{t \in \mathcal{T}_h} B_{m,t}^{\mathrm{TGC}}, \quad \forall m \in \mathcal{M} \tag{16}$$

This formulation links carbon accounting to the microgrid dispatch variables that directly enter the power-balance model. The TGCs term is incorporated through its carbon-equivalent entitlement contribution, while the corresponding procurement cost is defined together with the other objective components in the system-level operating objective.

3.4. System-level operating cost and scalarized objective

A system-level operating objective is formulated to coordinate microgrid dispatch, data center spatiotemporal flexibility, SLA assurance, and carbon-related procurement. The objective consists of an economic operating cost, a carbon-related procurement cost, and an SLA penalty, which are retained as separate components before scalarization to preserve their respective operational interpretations.

The local generation and energy-storage operation costs are modeled as:

$$\begin{cases} C_m^{\mathrm{gen}}(P_{m,t}^{\mathrm{gen}}) = a_m (P_{m,t}^{\mathrm{gen}})^2 + b_m P_{m,t}^{\mathrm{gen}} + c_m \\ C_m^{\mathrm{ess}}(P_{m,t}^{\mathrm{ch}}, P_{m,t}^{\mathrm{dis}}) = c_m^{\mathrm{deg}} \left(P_{m,t}^{\mathrm{ch}} + P_{m,t}^{\mathrm{dis}} \right) \\ \forall m \in \mathcal{M},\ t \in \mathcal{T}_h,\ a_m \geq 0 \end{cases} \tag{17}$$

The operating cost associated with data center spatiotemporal flexibility is defined as the sum of the net settlement for data center power deviations and the workload-migration cost:

$$\begin{aligned} C^{\mathrm{stf}} = & \sum_{t \in \mathcal{T}_h} \sum_{d \in \mathcal{D}} \left(\pi_t^{\mathrm{DC},+} p_{d,t}^{\mathrm{DC},+} - \pi_t^{\mathrm{DC},-} p_{d,t}^{\mathrm{DC},-} \right) \Delta t \\ & + \sum_{t \in \mathcal{T}_h} \sum_{m \in \mathcal{M}} \sum_{d \in \mathcal{D}_m} \sum_{d' \in \mathcal{D} \setminus \{d\}} \left(c_{\mathrm{int}}^{\mathrm{mig}} v_{d \to d',t} + c_{\mathrm{bat}}^{\mathrm{mig}} \sum_{\tau=t}^{t+\mathcal{W}_{d,t}^{\mathrm{eff}}} q_{d \to d',t \to \tau} \right) \end{aligned} \tag{18}$$

The net settlement of data center power deviations from the baseline is represented by the first term. The prices are assumed to satisfy $\pi_t^{\mathrm{DC},+} > \pi_t^{\mathrm{DC},-} \geq 0$ for all $t \in \mathcal{T}_h$. Under this condition, simultaneous positive deviation components are excluded at optimality, because both can be reduced without changing the net deviation while decreasing C^{stf}. Power deviations are converted into energy quantities through multiplication by Δt. The second term represents the workload-migration cost.

The SLA penalty for unserved workloads is:

$$C^{\mathrm{SLA}} = \sum_{t \in \mathcal{T}_h} \sum_{d \in \mathcal{D}} \left(\pi_{\mathrm{int}}^{\mathrm{drop}} w_{d,t}^{\mathrm{drop,int}} + \pi_{\mathrm{bat}}^{\mathrm{drop}} w_{d,t}^{\mathrm{drop,bat}} \right) \tag{19}$$

The penalty coefficients satisfy $\pi_{\mathrm{int}}^{\mathrm{drop}} > \pi_{\mathrm{bat}}^{\mathrm{drop}} \geq 0$, reflecting the stricter service requirement of interactive workloads.

The carbon-related procurement cost associated with carbon emission rights and TGCs is:

$$C^{\mathrm{carbon}} = \sum_{t \in \mathcal{T}_h} \sum_{m \in \mathcal{M}} \left(\pi_t^{\mathrm{CER}} B_{m,t}^{\mathrm{CER}} + \pi_t^{\mathrm{TGC}} B_{m,t}^{\mathrm{TGC}} \right) \tag{20}$$

The economic operating cost is obtained by combining grid transactions, local device operation, and spatiotemporal-flexibility-related operating cost:

$$\begin{aligned} C^{\mathrm{econ}} = & \sum_{t \in \mathcal{T}_h} \sum_{m \in \mathcal{M}} \Big(\pi_t^{\mathrm{buy}} P_{m,t}^{\mathrm{buy}} - \pi_t^{\mathrm{sell}} P_{m,t}^{\mathrm{sell}} + C_m^{\mathrm{gen}}(P_{m,t}^{\mathrm{gen}}) \\ & + C_m^{\mathrm{ess}}(P_{m,t}^{\mathrm{ch}}, P_{m,t}^{\mathrm{dis}}) \Big) \Delta t + C^{\mathrm{stf}} \end{aligned} \tag{21}$$

The scalarized operating objective is formulated as:

$$\min_{\Xi} \ J := C^{\mathrm{econ}} + w^{\mathrm{carbon}} C^{\mathrm{carbon}} + w^{\mathrm{sla}} C^{\mathrm{SLA}} \tag{22}$$

Here, Ξ denotes the collection of all scheduling decision variables constrained by the preceding microgrid, data center, and carbon-accounting models. The weighting coefficients w^{carbon} and w^{sla} are used to scalarize the carbon-related procurement cost and the SLA penalty with the economic operating cost.

4. Uncertainty modeling and wasserstein-DRCC reformulation

The coordinated optimization model in Section 3 is affected by uncertainties in renewable generation, local non-data center load, and data center workload arrivals. These uncertainties couple power dispatch with data center service assurance and therefore require explicit risk control. A Wasserstein-DRCC framework is developed to model distributional ambiguity from finite samples without prescribing a parametric probability distribution. This section defines the joint uncertainty representation and Wasserstein ambiguity set, formulates the critical operational constraints under distributional robustness, and presents a tractable distributionally robust Conditional Value-at-Risk (DR-CVaR) reformulation with constraint screening.

4.1. Multisource uncertainty representation and wasserstein ambiguity set

At each period t, multisource uncertainty is represented by a joint vector containing local non-data center load, renewable generation, and data center workload arrivals. The uncertainty vector is defined as:

$$\boldsymbol{\xi}_t := \left[\{\tilde{P}_{m,t}^{\mathrm{load}}\}_{m \in \mathcal{M}}, \{\tilde{P}_{m,t}^{\mathrm{ren}}\}_{m \in \mathcal{M}}, \{\tilde{\lambda}_{d,t}^{\mathrm{int}}\}_{d \in \mathcal{D}}, \{\tilde{\lambda}_{d,t}^{\mathrm{bat}}\}_{d \in \mathcal{D}} \right]^{\top} \in \mathbb{R}^{n_\xi} \tag{23}$$

The random variables in $\boldsymbol{\xi}_t$ are the uncertain counterparts of the corresponding quantities in the coordinated optimization model of

Section 3. They represent the uncertainty sources that directly affect power dispatch and workload scheduling, namely local non-data center load, renewable generation, interactive workload arrivals, and batch workload arrivals. Parameters not included in ξ_t are treated as deterministic in the present formulation.

Each sample $\hat{\xi}_t^{(n)}$ is a synchronized observation of all components in period t. Hence, cross-source dependence is preserved in the empirical distribution without imposing independence among uncertainty sources. The samples are expressed in scaled coordinates for Wasserstein-distance evaluation.

Given N_t available samples $\{\hat{\xi}_t^{(n)}\}_{n=1}^{N_t}$, the empirical distribution at period t is defined as:

$$\hat{\mathbb{P}}_t := (1/N_t)\sum_{n=1}^{N_t}\delta_{\hat{\xi}_t^{(n)}} \tag{24}$$

Here, $\delta_{\hat{\xi}_t^{(n)}}$ denotes the Dirac measure at $\hat{\xi}_t^{(n)}$. The period-wise empirical distributions allow time-varying uncertainty statistics to be represented, while interperiod coupling is retained through the storage dynamics and workload-deferral constraints in Section 3.

Distributional ambiguity around $\hat{\mathbb{P}}_t$ is described by the following Wasserstein ambiguity set:

$$\mathcal{P}_t(\rho_t) := \left\{\mathbb{P}\in\mathcal{P}(\mathcal{X}_t) \mid W_1(\mathbb{P},\hat{\mathbb{P}}_t)\le\rho_t\right\} \tag{25}$$

Here, $\mathcal{P}(\mathcal{X}_t)$ is the set of probability distributions supported on $\mathcal{X}_t\subseteq\mathbb{R}^{n_\xi}$, and $W_1(\mathbb{P},\hat{\mathbb{P}}_t)$ is the first-order Wasserstein distance. The radius ρ_t specifies the admissible distributional deviation from $\hat{\mathbb{P}}_t$; increasing ρ_t enlarges the ambiguity set and yields a more conservative formulation.

4.2. Wasserstein-DRCC formulation of critical operational constraints

The Wasserstein ambiguity set $\mathcal{P}_t(\rho_t)$ is used to impose risk control on constraints associated with power-balance feasibility and data center service assurance. In the proposed formulation, the risk-controlled set is restricted to two critical operational constraints: microgrid power balance and interactive-workload service coverage.

For microgrid power balance, uncertainty enters through local non-data center load and renewable generation, represented by $\tilde{P}_{m,t}^{\text{load}}$ and $\tilde{P}_{m,t}^{\text{ren}}$, respectively. The corresponding net-supply violation function is defined as:

$$\begin{aligned} g_{m,t}^{\text{bal}}(\Xi,\xi_t) := (\tilde{P}_{m,t}^{\text{load}}-\tilde{P}_{m,t}^{\text{ren}}) + \sum_{d\in\mathcal{D}_m}P_{d,t}^{\text{DC}} - \Big(P_{m,t}^{\text{gen}}+P_{m,t}^{\text{dis}} \\ + P_{m,t}^{\text{buy}}+P_{m,t}^{\text{in}}-P_{m,t}^{\text{ch}}-P_{m,t}^{\text{sell}}-P_{m,t}^{\text{out}}\Big) \end{aligned} \tag{26}$$

The condition $g_{m,t}^{\text{bal}}(\Xi,\xi_t)\le 0$ indicates that the realized power-balance requirement of microgrid m is satisfied under the scheduled dispatch. A positive value corresponds to a net-supply shortage.

For interactive workloads, service coverage must be ensured within the arrival period. With uncertain interactive workload arrival $\tilde{\lambda}_{d,t}^{\text{int}}$, the service-coverage violation function is defined as:

$$g_{d,t}^{\text{sla}}(\Xi,\xi_t) := \tilde{\lambda}_{d,t}^{\text{int}} - \sum_{d'\in\mathcal{D}}\nu_{d\to d',t} \tag{27}$$

The condition $g_{d,t}^{\text{sla}}(\Xi,\xi_t)\le 0$ ensures that the scheduled same-period service commitment covers the realized interactive workload. Since $w_{d,t}^{\text{drop,int}}$ represents unserved workload, it is excluded from service coverage and penalized separately. Batch workloads are handled through admissible deferral and unserved-workload penalties rather than risk-controlled constraints.

The corresponding risk-controlled index set is defined as:

$$\mathcal{I} = \{(m,\text{bal}) \mid m\in\mathcal{M}\}\cup\{(d,\text{sla}) \mid d\in\mathcal{D}\} \tag{28}$$

For each $i\in\mathcal{I}$ and $t\in\mathcal{T}_h$, the Wasserstein-DRCC is imposed as:

$$\inf_{\mathbb{P}\in\mathcal{P}_t(\rho_t)}\mathbb{P}\left(g_{i,t}(\Xi,\xi_t)\le 0\right)\ge 1-\varepsilon_{i,t} \tag{29}$$

Here, $\varepsilon_{i,t}$ denotes the prescribed violation probability. The inequality enforces a worst-case satisfaction probability of at least $1-\varepsilon_{i,t}$ over all distributions in $\mathcal{P}_t(\rho_t)$.

4.3. Tractable DR-CVaR reformulation and constraint screening

The Wasserstein-DRCC in Eq. (29) is infinite-dimensional because the underlying distribution is optimized over the ambiguity set $\mathcal{P}_t(\rho_t)$. A tractable conservative reformulation is obtained through a DR-CVaR surrogate. For each $(i,t)\in\mathcal{I}\times\mathcal{T}_h$, the risk-controlled constraint function is affine in ξ_t:

$$g_{i,t}(\Xi,\xi_t) = a_{i,t}(\Xi)+\mathbf{b}_{i,t}^{\top}\xi_t \tag{30}$$

Here, $a_{i,t}(\Xi)$ collects the deterministic decision-dependent terms, and $\mathbf{b}_{i,t}$ is the coefficient vector associated with ξ_t. The affine dependence on uncertainty is retained because the server-count parameter $\bar{S}_{d,h}$ is fixed before the quasi-real-time problem is solved.

With $[x]_+=\max\{x,0\}$, the DR-CVaR surrogate is written as:

$$\inf_{\theta_{i,t}\in\mathbb{R}}\left\{a_{i,t}(\Xi)+\theta_{i,t}+(1/\varepsilon_{i,t})\sup_{\mathbb{P}\in\mathcal{P}_t(\rho_t)}\mathbb{E}_{\mathbb{P}}\left[(\mathbf{b}_{i,t}^{\top}\xi_t-\theta_{i,t})_+\right]\right\}\le 0 \tag{31}$$

Under the L_1-induced Wasserstein metric, the dual norm is $\|\cdot\|_\infty$. Hence, Eq. (31) can be conservatively reformulated as the following finite-dimensional convex constraints:

$$\begin{cases} a_{i,t}(\Xi)+\theta_{i,t}+\sum_{n=1}^{N_t}s_{i,t}^{(n)}/(\varepsilon_{i,t}N_t)+\rho_t\|\mathbf{b}_{i,t}\|_\infty/\varepsilon_{i,t}\le 0 \\ \mathbf{b}_{i,t}^{\top}\hat{\xi}_t^{(n)}-\theta_{i,t}\le s_{i,t}^{(n)} \\ s_{i,t}^{(n)}\ge 0 \\ n=1,\dots,N_t \end{cases} \tag{32}$$

The variable $\theta_{i,t}$ is the CVaR threshold, and $s_{i,t}^{(n)}$ denotes the excess of the n-th sample above this threshold. The last term in the first inequality is the Wasserstein regularization term induced by distributional ambiguity.

For the power-balance constraint, the deterministic term is:

$$\begin{aligned} a_{m,t}^{\text{bal}}(\Xi) = \sum_{d\in\mathcal{D}_m}P_{d,t}^{\text{DC}} - \Big(P_{m,t}^{\text{gen}}+P_{m,t}^{\text{dis}}+P_{m,t}^{\text{buy}}+P_{m,t}^{\text{in}} \\ - P_{m,t}^{\text{ch}}-P_{m,t}^{\text{sell}}-P_{m,t}^{\text{out}}\Big) \end{aligned} \tag{33}$$

The corresponding finite-sample DR-CVaR counterpart is:

$$\begin{cases} a_{m,t}^{\text{bal}}(\Xi)+\theta_{m,t}^{\text{bal}}+\sum_{n=1}^{N_t}s_{m,t}^{\text{bal},(n)}/(\varepsilon_{m,t}^{\text{bal}}N_t)+\rho_t\|\mathbf{b}_{m,t}^{\text{bal}}\|_\infty/\varepsilon_{m,t}^{\text{bal}}\le 0 \\ \hat{P}_{m,t}^{\text{load},(n)}-\hat{P}_{m,t}^{\text{ren},(n)}-\theta_{m,t}^{\text{bal}}\le s_{m,t}^{\text{bal},(n)} \\ s_{m,t}^{\text{bal},(n)}\ge 0 \\ n=1,\dots,N_t \end{cases} \tag{34}$$

For the interactive-SLA constraint, the deterministic term is:

$$a_{d,t}^{\text{sla}}(\Xi) := -\sum_{d'\in\mathcal{D}}\nu_{d\to d',t} \tag{35}$$

The corresponding finite-sample DR-CVaR counterpart is:

$$\begin{cases} a^{\mathrm{sla}}_{d,t}(\Xi) + \theta^{\mathrm{sla}}_{d,t} + \sum_{n=1}^{N_t} s^{\mathrm{sla},(n)}_{d,t}/(\varepsilon^{\mathrm{sla}}_{d,t} N_t) + \rho_t \|\mathbf{b}^{\mathrm{sla}}_{d,t}\|_\infty/\varepsilon^{\mathrm{sla}}_{d,t} \le 0 \\ \hat{\lambda}^{\mathrm{int},(n)}_{d,t} - \theta^{\mathrm{sla}}_{d,t} \le s^{\mathrm{sla},(n)}_{d,t} \\ s^{\mathrm{sla},(n)}_{d,t} \ge 0 \\ n = 1, \dots, N_t \end{cases} \tag{36}$$

Eq. (32) introduces N_t sample-excess variables for each risk-controlled constraint-time pair, so the auxiliary-variable dimension scales with the sample size and the number of candidate pairs. A constraint screening rule is therefore introduced to reduce the size of the reformulated problem.

For a given solution Ξ, the screening indicator is defined as:

$$\zeta_{i,t}(\Xi) := \max_{n=1,\dots,N_t} g_{i,t}\big(\Xi, \hat{\xi}^{(n)}_t\big) + \rho_t \|\mathbf{b}_{i,t}\|_\infty/\varepsilon_{i,t} \tag{37}$$

Given a safety margin $\delta_{i,t} \ge 0$, the effective constraint-time pair set is constructed from the screening indicator evaluated at a reference solution $\Xi^{(0)}$:

$$\mathcal{I}^{\mathrm{eff}}_h := \big\{(i,t) \in \mathcal{I} \times \mathcal{T}_h \mid \zeta_{i,t}(\Xi^{(0)}) > -\delta_{i,t}\big\} \tag{38}$$

The DR-CVaR constraints are imposed on the pairs in $\mathcal{I}^{\mathrm{eff}}_h$. After solving the reduced problem, the screening indicator is re-evaluated for all pairs in $\mathcal{I} \times \mathcal{T}_h$ at the updated solution. Any omitted pair that fails the screening test is added to $\mathcal{I}^{\mathrm{eff}}_h$, and the reduced problem is resolved until no additional pair is activated. This screening-validation procedure reduces the number of auxiliary variables while retaining an explicit feasibility check over all risk-controlled constraint-time pairs.

5. SAC-based day-ahead server capacity provisioning

The fast-timescale Wasserstein-DRCC scheduling model is formulated with a fixed server-count parameter. If server counts were optimized jointly with workload scheduling and microgrid dispatch, mixed-integer coupling would be introduced into the data center capacity and power models, which would weaken the tractability of the reformulated scheduling problem. To avoid this coupling, server capacity is provisioned at the day-ahead layer through SAC and is passed to the fast-timescale model as a fixed parameter. This section defines the role of server provisioning in the bi-timescale framework, formulates the corresponding decision process and reward feedback, and presents the SAC training procedure for integer server-count actions. The resulting bi-timescale scheduling workflow is summarized in Fig. 3.

5.1. Role of server provisioning in the bi-timescale framework

For each data center, the server-count decision is generated by the SAC layer before the fast-timescale scheduling problem over $\mathcal{T}_h$ is solved. The resulting server-count decision is then fixed in the fast-timescale model as:

$$\bar{S}_{d,h} := S_{d,h}, \quad \forall d \in D,\ h \in \mathcal{T}^{\mathrm{slow}} \tag{39}$$

The overbar indicates that the day-ahead server-count decision is fixed when the fast-timescale model is solved, rather than representing an upper bound or an average value. For all $t \in \mathcal{T}_h$, $\bar{S}_{d,h}$ is treated as a known parameter in the workload-capacity constraint and the data center power model.

Through $\bar{S}_{d,h}$, server provisioning constrains the service capability of each data center and shapes its power demand. On the computing side, the fixed server count determines the available processing capacity within each scheduling period. On the power side, it contributes to the idle information technology power and thereby affects data center power consumption, microgrid dispatch, and carbon-emission accounting through the resulting operating decisions.

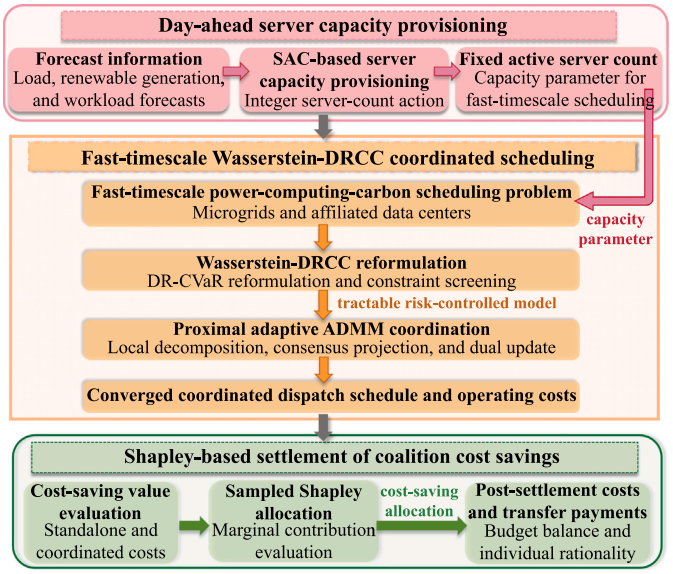


Fig. 3. Bi-timescale scheduling and settlement framework.

This design separates the integer server-count decision from the continuous fast-timescale scheduling variables. Accordingly, the fast-timescale Wasserstein-DRCC scheduling model treats $\bar{S}_{d,h}$ as fixed within each horizon, while the effect of the selected server configuration is evaluated through the solved operating schedule. This separation provides the modeling basis for coordinating server provisioning with SLA-oriented workload scheduling and power-computing-carbon operations without embedding integer server-count variables in the fast-timescale robust optimization model.

5.2. Markov decision process and cross-timescale reward

At the beginning of each slow-timescale period $h \in \mathcal{T}^{\mathrm{slow}}$, predictive information for the fast-timescale horizon $\mathcal{T}_h$ is observed by the SAC layer, and a server-count action is determined. The selected server count is fixed as $\bar{S}_{d,h}$ before the fast-timescale Wasserstein-DRCC scheduling problem is solved. The server-provisioning problem is formulated as a Markov decision process (MDP), whose state, action, transition, and reward are specified as follows.

The state vector consists of predictive statistics of workload arrivals, renewable generation, and local non-data center load over $\mathcal{T}_h$:

$$\mathbf{s}_h = \Big[\{\hat{\mu}^{\mathrm{int}}_{d,h}, \hat{\sigma}^{\mathrm{int}}_{d,h}, \hat{\mu}^{\mathrm{bat}}_{d,h}, \hat{\sigma}^{\mathrm{bat}}_{d,h}\}_{d\in D}, \{\hat{\mu}^{\mathrm{ren}}_{m,h}, \hat{\sigma}^{\mathrm{ren}}_{m,h}, \hat{\mu}^{\mathrm{load}}_{m,h}, \hat{\sigma}^{\mathrm{load}}_{m,h}\}_{m\in\mathcal{M}}\Big]^{\top} \in \mathbb{R}^{n_s} \tag{40}$$

Here, n_s denotes the dimension of the state vector. These statistics are derived from the uncertainty samples $\{\hat{\xi}^{(n)}_t\}_{n=1}^{N_t}$ for $t \in \mathcal{T}_h$, consistent with the uncertainty representation in Section 4.1. The state includes only exogenous predictive information and excludes fast-timescale decision variables.

The action is the integer-valued server-count vector:

$$\mathbf{a}_h = (S_{d,h})_{d\in D}, \quad S_{d,h} \in \{S^{\min}_d, \dots, S^{\max}_d\}, \quad \forall d \in D \tag{41}$$

After $\mathbf{a}_h$ is selected, $S_{d,h}$ is passed to the fast-timescale model as $\bar{S}_{d,h}$ according to Eq. (39) and is used for all $t \in \mathcal{T}_h$. The continuous relaxation and integer mapping adopted for SAC training are presented in Section 5.3.

The state transition is driven by the predictive statistics of the next fast-timescale horizon:

$$\mathbf{s}_{h+1} = \boldsymbol{\omega}_{h+1} \tag{42}$$

Here, $\boldsymbol{\omega}_{h+1}$ denotes the predictive-statistics vector over $\mathcal{T}_{h+1}$. The server-count action affects the operating schedule within the current fast-timescale horizon through $\bar{S}_{d,h}$, but it is not assumed to alter the exogenous uncertainty statistics of the next horizon. Therefore, its effect is reflected in the reward rather than in the state-transition model.

When a feasible fast-timescale schedule Ξ^* is obtained under the fixed server configuration $\bar{S}_{d,h}$, the operating-cost feedback $\hat{C}_h^{\mathrm{op}}$ is evaluated from the economic and carbon-related components of the scalarized objective, with the SLA penalty retained as a separate reward term:

$$\hat{C}_h^{\mathrm{op}} = \left(C^{\mathrm{econ}} + w^{\mathrm{carbon}} C^{\mathrm{carbon}}\right)\Big|_{\Xi=\Xi^*} \tag{43}$$

Here, C^{econ} comprises grid transactions, local device operation, and data center spatiotemporal-flexibility cost, whereas C^{carbon} accounts for the procurement cost of carbon emission rights and TGCs.

The feasibility status of the fast-timescale Wasserstein-DRCC problem is encoded as:

$$\Phi_h^{\mathrm{DRCC}} = \begin{cases} 0, & \text{feasible} \\ \Phi^{\max}, & \text{infeasible} \end{cases} \qquad \Phi^{\max} > 0 \tag{44}$$

Here, $\theta_{i,t}^*$ and $s_{i,t}^{(n),*}$ remain auxiliary variables of the fast-timescale solution.

For the same feasible solution, the SLA penalty feedback is:

$$\hat{C}_h^{\mathrm{SLA}} = C^{\mathrm{SLA}}\Big|_{\Xi=\Xi^*} \tag{45}$$

For a feasible fast-timescale solution, the instantaneous reward is defined as:

$$r_h = -\psi^{\mathrm{econ}} \hat{C}_h^{\mathrm{op}} - \psi^{\mathrm{SLA}} \hat{C}_h^{\mathrm{SLA}} \tag{46}$$

If no feasible fast-timescale solution is obtained, the reward is defined as:

$$r_h = -\psi^{\mathrm{infeas}} \Phi_h^{\mathrm{DRCC}} = -\psi^{\mathrm{infeas}} \Phi^{\max} \tag{47}$$

The reward weights are scaled as:

$$\begin{cases} \psi^{\mathrm{econ}} = 1 \\ \psi^{\mathrm{infeas}} = \mathbb{E}[\hat{C}_h^{\mathrm{op}}] / \left(\mathbb{E}[\Phi_h^{\mathrm{DRCC}}] + \epsilon_0\right) \\ \psi^{\mathrm{SLA}} = \mathbb{E}[\hat{C}_h^{\mathrm{op}}] / \left(\mathbb{E}[\hat{C}_h^{\mathrm{SLA}}] + \epsilon_0\right) \end{cases} \tag{48}$$

For reward scaling, the cost and SLA averages are defined over feasible transitions, whereas $\mathbb{E}[\Phi_h^{\mathrm{DRCC}}]$ is defined over all sampled server configurations. The constant ϵ_0 prevents division by zero. Feasible configurations are evaluated by cost and SLA performance, while infeasible configurations are penalized by $\Phi^{\max}$.

5.3. SAC-based policy learning for integer server provisioning

Based on the Markov decision process defined in Section 5.2, SAC is used to learn a day-ahead server-provisioning policy. The policy determines the integer server-count vector $\mathbf{a}_h = (S_{d,h})_{d\in\mathcal{D}}$, which is passed to the fast-timescale scheduling model as the fixed parameter $\bar{S}_{d,h}$. To support policy optimization with integer server-count actions, a continuous latent action is sampled and then mapped to the admissible integer action space.

The maximum-entropy objective is:

$$J(\pi_\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{h=0}^{|\mathcal{T}^{\mathrm{slow}}|-1} \gamma^h \left(r_h - \alpha \log \pi_\theta(\mathbf{z}_h|\mathbf{s}_h)\right)\right] \tag{49}$$

where $\mathbf{z}_h \in \mathbb{R}^{|\mathcal{D}|}$ denotes the continuous latent action, $\mathbb{E}_{\pi_\theta}[\cdot]$ is the expectation under policy π_θ, $\gamma \in (0,1)$ is the discount factor, and $\alpha > 0$ is the entropy temperature.

The continuous latent action is sampled by the reparameterization form:

$$\mathbf{z}_h = \boldsymbol{\mu}_\theta(\mathbf{s}_h) + \boldsymbol{\sigma}_\theta(\mathbf{s}_h) \odot \boldsymbol{\eta}_h, \quad \boldsymbol{\eta}_h \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{|\mathcal{D}|}) \tag{50}$$

The sampled latent action is then mapped to an admissible integer server-count action:

$$S_{d,h} = \mathrm{clip}\Big(\mathrm{round}\big(S_d^{\min} + (S_d^{\max} - S_d^{\min})\sigma(z_{d,h})\big), S_d^{\min}, S_d^{\max}\Big) \tag{51}$$

The resulting integer vector $\mathbf{a}_h = (S_{d,h})_{d\in\mathcal{D}}$ is fixed as $\bar{S}_{d,h}$ in the fast-timescale scheduling problem.

Because $\mathrm{round}(\cdot)$ is non-differentiable, the Straight-Through Estimator (STE) is used for backpropagation. The surrogate derivative is:

$$\partial S_{d,h} / \partial z_{d,h} \approx (S_d^{\max} - S_d^{\min})\sigma(z_{d,h})\big(1 - \sigma(z_{d,h})\big) \tag{52}$$

Thus, the forward pass uses the integer server-count action, while the backward pass propagates gradients through the differentiable surrogate.

Let $\mathcal{R}$ denote the replay buffer storing transitions $(\mathbf{s}_h, \mathbf{a}_h, r_h, \mathbf{s}_{h+1})$. For a non-terminal transition, the temporal-difference target is:

$$y_h = r_h + \gamma\Big(\min_{j=1,2} Q_{\bar{\phi}_j}(\mathbf{s}_{h+1}, \mathbf{a}'_{h+1}) - \alpha \log \pi_\theta(\mathbf{z}'_{h+1}|\mathbf{s}_{h+1})\Big) \tag{53}$$

where $\mathbf{z}'_{h+1} \sim \pi_\theta(\cdot|\mathbf{s}_{h+1})$, and $\mathbf{a}'_{h+1}$ is obtained from $\mathbf{z}'_{h+1}$ by Eq. (51).

The twin critics are updated by minimizing:

$$\mathcal{L}(\phi_j) = \mathbb{E}_{\mathcal{R}}\left[\left(Q_{\phi_j}(\mathbf{s}_h, \mathbf{a}_h) - y_h\right)^2\right], \quad j = 1,2 \tag{54}$$

The policy is updated by minimizing:

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathbf{s}_h\sim\mathcal{R},\, \mathbf{z}_h\sim\pi_\theta(\cdot|\mathbf{s}_h)}\Big[\alpha \log \pi_\theta(\mathbf{z}_h|\mathbf{s}_h) - \min_{j=1,2} Q_{\phi_j}(\mathbf{s}_h, \mathbf{a}_h)\Big] \tag{55}$$

where $\mathbf{a}_h$ is obtained from $\mathbf{z}_h$ by Eq. (51). The integer-mapping step ensures that the scheduling model receives only admissible server-count actions.

The entropy temperature is updated by minimizing:

$$\mathcal{L}(\alpha) = \mathbb{E}_{\mathbf{s}_h\sim\mathcal{R},\, \mathbf{z}_h\sim\pi_\theta(\cdot|\mathbf{s}_h)}\left[-\alpha\left(\log \pi_\theta(\mathbf{z}_h|\mathbf{s}_h) + \mathcal{H}^{\mathrm{tar}}\right)\right] \tag{56}$$

The target entropy is set as $\mathcal{H}^{\mathrm{tar}} = -|\mathcal{D}|$, consistent with the dimension of the latent action.

The target critics are updated by soft replacement:

$$\bar{\phi}_j \leftarrow \tau_{\mathrm{tar}} \phi_j + (1 - \tau_{\mathrm{tar}})\bar{\phi}_j, \quad j = 1,2 \tag{57}$$

where $\tau_{\mathrm{tar}} \in (0,1)$ is the target-network update coefficient. Under this training scheme, each server-provisioning action is assessed through the reward feedback defined in Section 5.2, while the fast-timescale scheduling model receives only the mapped integer server-count action.

6. Distributed solution and coalition settlement

The fast-timescale Wasserstein-DRCC scheduling problem couples microgrid dispatch, data center workload scheduling, carbon-related procurement, and SLA assurance across multiple microgrids. To avoid centralized disclosure of local models, cost functions, device constraints, data center parameters, and risk-control constraints, the reformulated scheduling problem is decomposed by microgrid and solved through a proximal adaptive ADMM. At convergence, the mutual-support components of the ADMM dual vectors are interpreted as risk-aware price signals for coordinated power exchange. The cooperative cost saving obtained from coordinated operation is then allocated through a sampled Shapley-based rule. This section presents the local problem decomposition, the proximal adaptive ADMM solution, risk-aware pricing with privacy-preserving information exchange, and coalition cost-saving settlement.

6.1. Local decomposition of the wasserstein-DRCC scheduling problem

Given the fixed server-count parameter $\bar{S}_{d,h}$, the reformulated Wasserstein-DRCC scheduling problem over $\mathcal{T}_h$ is organized into microgrid-wise local subproblems. Each local subproblem contains the operating variables of one microgrid, the scheduling variables of its associated data centers, and the DR-CVaR auxiliary variables assigned to its risk-controlled constraints. For microgrid $m \in \mathcal{M}$, the local decision-variable set is defined as:

$$\Xi_m := \Xi_m^{\mathrm{MG}} \cup \Xi_m^{\mathrm{DC}} \cup \Xi_m^{\mathrm{DRCC}} \tag{58}$$

The dispatch, storage, grid-transaction, mutual-support, and carbon-related procurement variables of microgrid m are included in Ξ_m^{MG}, whereas the workload-scheduling and data-center power variables associated with $\mathcal{D}_m$ are included in Ξ_m^{DC}. For each cross-microgrid workload assignment, the associated variables are represented locally at both the source and destination microgrids, and consistency between the two local representations is enforced through the ADMM consensus mechanism.

For $d \in \mathcal{D}_m$ and $d' \in \mathcal{D}_{m'}$ with $m \neq m'$, the source-side and destination-side local representations are denoted by $v^{m}_{d\to d',t}$ and $v^{m'}_{d\to d',t}$ for interactive workloads, and by $q^{m}_{d\to d',t\to\tau}$ and $q^{m'}_{d\to d',t\to\tau}$ for batch workloads, respectively.

The effective risk-controlled pairs associated with microgrid m are:

$$\begin{aligned}\mathcal{I}^{\mathrm{eff}}_{m,h} := \big\{(i,t) \in \mathcal{I}^{\mathrm{eff}}_{h} \mid\ & i = (m,\mathrm{bal}) \\ & \text{or } i = (d,\mathrm{sla}),\ d \in \mathcal{D}_m\big\}\end{aligned} \tag{59}$$

The corresponding local DR-CVaR auxiliary-variable set is:

$$\begin{aligned}\Xi_m^{\mathrm{DRCC}} := & \left\{\theta_{i,t} \mid (i,t) \in \mathcal{I}^{\mathrm{eff}}_{m,h}\right\} \\ & \cup \left\{s^{(n)}_{i,t} \mid (i,t) \in \mathcal{I}^{\mathrm{eff}}_{m,h},\ n = 1,\dots,N_t\right\}\end{aligned} \tag{60}$$

The local feasible set Ω_m contains the deterministic operating constraints associated with microgrid m and the data centers in $\mathcal{D}_m$, the relevant local representations of cross-microgrid workload assignments, and the finite-dimensional DR-CVaR constraints in Section 4.3 for all $(i,t) \in \mathcal{I}^{\mathrm{eff}}_{m,h}$. Under this decomposition, the local scalarized objective is defined as:

$$J_m(\Xi_m) := C_m^{\mathrm{econ}} + w^{\mathrm{carbon}} C_m^{\mathrm{carbon}} + w^{\mathrm{sla}} C_m^{\mathrm{SLA}} \tag{61}$$

The local economic component C_m^{econ} is obtained by grouping the grid-transaction, local device-operation, data-center power-deviation, and workload-migration terms associated with microgrid m and the data centers in $\mathcal{D}_m$. Accordingly, the system-level scalarized objective in Eq. (22) is decomposed as:

$$J = \sum_{m\in\mathcal{M}} J_m(\Xi_m) \tag{62}$$

The net mutual-support injection of microgrid m is defined as:

$$u_{m,t}(\Xi_m) := P^{\mathrm{out}}_{m,t} - P^{\mathrm{in}}_{m,t}, \quad \forall m \in \mathcal{M},\ t \in \mathcal{T}_h \tag{63}$$

The net mutual-support injections and local representations of cross-microgrid workload assignments are collected in the boundary vector $\mathbf{u}_m(\Xi_m)$, and $\mathbf{v}_m$ denotes the corresponding consensus vector. Their stacked forms are denoted by $\mathbf{u}(\Xi)$ and $\mathbf{v}$, respectively. The set C enforces mutual-support balance and cross-microgrid workload consistency. The decomposed consensus form is:

$$\begin{aligned}\min_{\{\Xi_m\}_{m\in\mathcal{M}},\,\mathbf{v}} \quad & \sum_{m\in\mathcal{M}} J_m(\Xi_m) \\ \text{s.t.} \quad & \Xi_m \in \Omega_m, \quad \forall m \in \mathcal{M} \\ & \mathbf{u}_m(\Xi_m) - \mathbf{v}_m = \mathbf{0}, \quad \forall m \in \mathcal{M} \\ & \mathbf{v} \in C\end{aligned} \tag{64}$$

Here, $\mathbf{u}$ and $\mathbf{v}$ stack the local boundary and consensus vectors, respectively. The set C enforces mutual-support balance and cross-microgrid workload consistency; once consensus is reached, the workload assignments correspond to the original variables in Section 3.

6.2. Proximal adaptive ADMM for distributed scheduling

Based on the consensus form in Eq. (64), the decomposed Wasserstein-DRCC scheduling problem is addressed through a proximal adaptive ADMM. The finite-dimensional DR-CVaR constraints are contained in the local feasible sets Ω_m, and therefore they are not repeated in the distributed updates.

Let $\boldsymbol{\varpi}_m$ denote the dual vector associated with $\mathbf{u}_m(\Xi_m) - \mathbf{v}_m = \mathbf{0}$, and let $\boldsymbol{\varpi}$ denote the stacked dual vector. The quantities $u_{m,t}(\Xi_m)$, $v_{m,t}$, and $\varpi_{m,t}$ are retained as the mutual-support components of $\mathbf{u}_m(\Xi_m)$, $\mathbf{v}_m$, and $\boldsymbol{\varpi}_m$, respectively. The ADMM penalty parameter at iteration k is denoted by $\beta^k > 0$. The augmented Lagrangian is:

$$\begin{aligned}\mathcal{L}_{\beta^k} = & \sum_{m\in\mathcal{M}} J_m(\Xi_m) + \sum_{m\in\mathcal{M}} \boldsymbol{\varpi}_m^{\mathsf{T}}\big(\mathbf{u}_m(\Xi_m) - \mathbf{v}_m\big) \\ & + (\beta^k/2) \sum_{m\in\mathcal{M}} \big\|\mathbf{u}_m(\Xi_m) - \mathbf{v}_m\big\|_2^2\end{aligned} \tag{65}$$

At iteration $k+1$, each microgrid solves the following local subproblem in parallel:

$$\begin{aligned}\Xi_m^{k+1} = \arg\min_{\Xi_m\in\Omega_m} \Big\{ & J_m(\Xi_m) + (\boldsymbol{\varpi}_m^k)^{\mathsf{T}}\big(\mathbf{u}_m(\Xi_m) - \mathbf{v}_m^k\big) \\ & + (\beta^k/2)\left\|\mathbf{u}_m(\Xi_m) - \mathbf{v}_m^k\right\|_2^2 + (\eta_m/2)\left\|\Xi_m - \Xi_m^k\right\|_2^2 \Big\}\end{aligned} \tag{66}$$

The proximal term regularizes the local update without changing Ω_m. The consensus vector is then updated by Euclidean projection onto C:

$$\mathbf{v}^{k+1} = \mathrm{Proj}_C\left(\mathbf{u}(\Xi^{k+1}) + \boldsymbol{\varpi}^k/\beta^k\right) \tag{67}$$

Here, $\mathbf{u}(\Xi^{k+1})$ stacks $\mathbf{u}_m(\Xi_m^{k+1})$ over all microgrids. Mutual-support balance and cross-microgrid workload consistency are enforced by this projection.

The dual vectors are updated as:

$$\boldsymbol{\varpi}_m^{k+1} = \boldsymbol{\varpi}_m^k + \beta^k \left(\mathbf{u}_m(\Xi_m^{k+1}) - \mathbf{v}_m^{k+1}\right), \quad \forall m \in \mathcal{M} \tag{68}$$

The primal and dual residuals are evaluated by:

$$\begin{cases} r_{\text{pri}}^{k+1} = \left\| \mathbf{u}(\Xi^{k+1}) - \mathbf{v}^{k+1} \right\|_2 \\ r_{\text{dual}}^{k+1} = \beta^k \left\| \mathbf{v}^{k+1} - \mathbf{v}^k \right\|_2 \end{cases} \tag{69}$$

The penalty parameter is adjusted by the residual-balancing rule:

$$\beta^{k+1} = \begin{cases} \tau_{\text{inc}}\beta^k, & r_{\text{pri}}^{k+1} > \mu_{\text{ad}} r_{\text{dual}}^{k+1} \\ \beta^k / \tau_{\text{dec}}, & r_{\text{dual}}^{k+1} > \mu_{\text{ad}} r_{\text{pri}}^{k+1} \\ \beta^k, & \text{otherwise} \end{cases} \tag{70}$$

The iterations are terminated when both residuals satisfy:

$$r_{\text{pri}}^{k+1} \le \epsilon_{\text{pri}}, \quad r_{\text{dual}}^{k+1} \le \epsilon_{\text{dual}} \tag{71}$$

At termination, each local solution satisfies the constraints in Ω_m, while mutual-support balance and cross-microgrid workload consistency are enforced within the prescribed ADMM tolerances. The converged mutual-support dual components $\varpi_{m,t}^*$ are used for risk-aware pricing.

6.3. Dual-based risk-aware pricing and privacy-preserving coordination

After ADMM convergence, $\varpi_{m,t}^*$, the mutual-support component of $\boldsymbol{\varpi}_m^*$, is used as a dual-informed price signal for mutual-support power exchange. It is associated with the constraint $u_{m,t}(\Xi_m) - v_{m,t} = 0$ and represents its marginal coordination value in the risk-controlled scheduling problem. The remaining dual components enforce cross-microgrid workload consistency and are not used for pricing.

Only boundary-level information is exchanged during the distributed iterations. At iteration k, the following information is transmitted by microgrid m:

$$\left\{\mathbf{u}_m(\Xi_m^{k+1}), \boldsymbol{\varpi}_m^k\right\} \tag{72}$$

After the consensus projection and dual update, the coordination layer returns:

$$\left\{\mathbf{v}_m^{k+1}, \boldsymbol{\varpi}_m^{k+1}\right\} \tag{73}$$

Local objective functions, device parameters, internal workload schedules, carbon-related procurement variables, and DR-CVaR auxiliary variables remain local. Only mutual-support quantities, local representations of cross-microgrid workload assignments, and the associated dual information are exchanged. Information exchange is therefore limited to the boundary variables required for distributed coordination.

The converged operating schedules, the resulting operating-cost components, and the dual-informed risk-aware price signals are then used for the coalition settlement mechanism in the next subsection.

6.4. Sampled shapley settlement for coalition cost savings

After the distributed Wasserstein-DRCC scheduling problem is solved by the improved ADMM developed above, the resulting coalition cost saving is allocated among microgrids according to their marginal contributions to the coalition value. The settlement is evaluated using the same scalarized operating objective as the scheduling model, so that the economic operating cost, the carbon-related procurement cost associated with carbon emission rights and TGCs, and the SLA penalty are accounted for consistently. The cost impact of data center workload scheduling is embedded in the local objective components of the affiliated microgrids.

Let Ξ_m^* denote the local schedule of microgrid m obtained after ADMM convergence. The coordinated local cost and the coordinated system cost are defined as:

$$J_m^* := J_m(\Xi_m^*), \quad J^* := \sum_{m\in\mathcal{M}} J_m^* \tag{74}$$

A non-cooperative baseline is used to define the standalone cost of each microgrid. In this baseline, coordinated interactions among microgrids and their affiliated data centers are disabled, and each microgrid is scheduled independently. The resulting standalone scalarized cost of microgrid m is denoted by $J_m^{(0)}$. The total standalone cost is:

$$J^{(0)} := \sum_{m\in\mathcal{M}} J_m^{(0)} \tag{75}$$

The total coalition cost saving is:

$$\Pi := J^{(0)} - J^* \tag{76}$$

For any coalition $S \subseteq \mathcal{M}$, the coalition value is defined as the cost saving obtained by the members of S:

$$\mathcal{V}(S) := \sum_{m\in S} J_m^{(0)} - J^*(S), \quad \mathcal{V}(\varnothing) = 0 \tag{77}$$

Here, $J^*(S)$ denotes the optimized scalarized cost of the microgrids in S when coordination is enabled only within S. With this definition, $\mathcal{V}(\mathcal{M}) = \Pi$.

The Shapley value of microgrid m is:

$$\phi_m = \sum_{S\subseteq\mathcal{M}\setminus\{m\}} \left(|S|!(|\mathcal{M}| - |S| - 1)!/|\mathcal{M}|!\right) \left[\mathcal{V}(S\cup\{m\}) - \mathcal{V}(S)\right], \quad \forall m \in \mathcal{M} \tag{78}$$

Because exact Shapley evaluation requires enumeration over all permutations of $\mathcal{M}$, a sampled-permutation approximation is adopted for computational tractability. Let $\{\sigma^r\}_{r=1}^K$ denote K sampled permutations of $\mathcal{M}$, and let $P_m^{\sigma^r}$ denote the set of microgrids preceding m in permutation σ^r. The sampled Shapley allocation is:

$$\hat{\phi}_m = (1/K)\sum_{r=1}^{K}\left[\mathcal{V}(P_m^{\sigma^r}\cup\{m\}) - \mathcal{V}(P_m^{\sigma^r})\right] \tag{79}$$

The post-settlement cost of microgrid m is:

$$J_m^{\text{sett}} := J_m^{(0)} - \hat{\phi}_m, \quad \forall m \in \mathcal{M} \tag{80}$$

When the same sampled permutations are used for all microgrids, budget balance is preserved as:

$$\sum_{m\in\mathcal{M}} \hat{\phi}_m = \Pi \tag{81}$$

Accordingly, the post-settlement costs satisfy the budget-balance condition:

$$\sum_{m\in\mathcal{M}} J_m^{\text{sett}} = J^{(0)} - \sum_{m\in\mathcal{M}} \hat{\phi}_m = J^* \tag{82}$$

Individual rationality is checked by:

$$J_m^{\text{sett}} \le J_m^{(0)}, \quad \forall m \in \mathcal{M} \tag{83}$$

According to Eq. (80), the individual-rationality condition is satisfied when $\hat{\phi}_m \ge 0$. The coalition cost saving is redistributed without changing the optimized total system cost. The settlement is therefore consistent with the risk-controlled operating objective used in the scheduling model.

Table 1
Case design for comparative analysis.

Case	Carbon mechanism	Spatial flexibility	Temporal flexibility	Uncertainty handling	Server provisioning
Case 1	✗	✗	✗	Deterministic	Fixed
Case 2	✗	✓	✗	Deterministic	Fixed
Case 3	✗	✓	✓	Deterministic	Fixed
Case 4	✓	✓	✓	Deterministic	Fixed
Case 5	✓	✓	✓	Scenario-SP	Fixed
Case 6	✓	✓	✓	W-DRCC	Fixed
Case 7	✓	✓	✓	W-DRCC	SAC-adaptive

7. Case studies

7.1. Test system and comparative case design

A test system with five microgrids and five data centers was used in the case study. Cross-microgrid scheduling is coordinated at the multi-microgrid level, and data center workload scheduling is coupled with microgrid operation through the proposed bi-timescale power-computing-carbon framework. The microgrids and data centers are configured with heterogeneous operating characteristics, including generation capacity, load scale, server configuration, PUE, SLA requirement, and workload composition. Based on this test system, the comparative case design is summarized in Table 1.

7.2. System-level outcomes and marginal cost contributions

Fig. 4 compares the daily operating cost, carbon emission, and SLA penalty across the seven cases. From Case 1 to Case 7, these three indicators were reduced by 32.3%, 27.7%, and 75.4%, respectively. The most favorable system-level outcome was achieved in Case 7, indicating that the proposed bi-timescale framework effectively coordinated data center workload flexibility with microgrid dispatch and the carbon mechanism. In the final coordinated case, the economic improvement was not obtained at the expense of emission performance or service reliability; instead, computing flexibility was converted into a schedulable resource for power-computing-carbon coordination in the multi-microgrid system.

The intermediate results in Fig. 4 further reveal the tradeoffs among economy, emission reduction, and service reliability. After the carbon mechanism was introduced in Case 4, carbon emission and daily operating cost were reduced, whereas the SLA penalty slightly increased relative to Case 3. This indicates that low-carbon dispatch alone may create a tradeoff with service quality if it is not jointly coordinated with data center workload scheduling. In Case 6, the Wasserstein distributionally robust chance constraint led to a small increase in daily operating cost compared with Case 5, but both carbon emission and SLA penalties were further reduced. This behavior is consistent with a limited robustness premium paid for improved reliability under uncertainty. When SAC-based server provisioning was further adopted in Case 7, the cost burden associated with fixed server provisioning was mitigated, and all three indicators reached their lowest levels.

Fig. 5 further quantifies the marginal cost contribution of each component. Spatial workload migration produced the largest cost reduction, confirming that geographical heterogeneity among microgrids and data centers is a major source of flexibility. Temporal workload reshaping provided an additional saving, showing its complementary role to spatial workload migration. The carbon mechanism and scenario-based stochastic programming also contributed smaller but positive cost reductions. By contrast, the Wasserstein distributionally robust chance constraint was the only component with a positive marginal cost contribution, reflecting its reliability-oriented conservatism. The final cost decrease from Case 6 to Case 7 confirms that SAC-based server provisioning is effective in recovering economic efficiency while preserving the robust quasi-real-time scheduling structure.

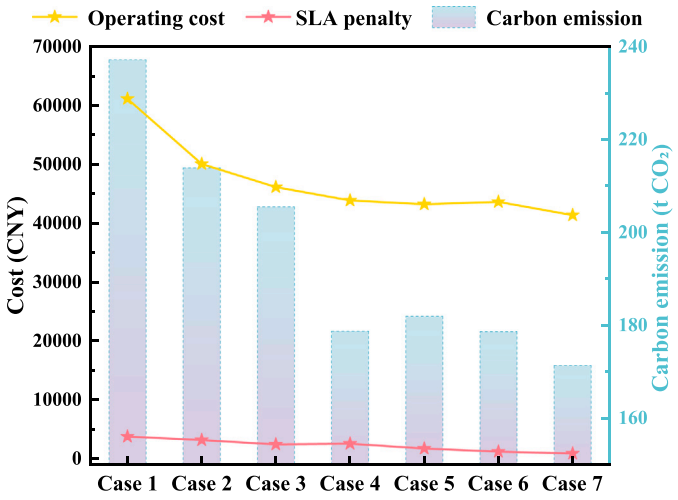


Fig. 4. Overall performance comparison across seven cases.

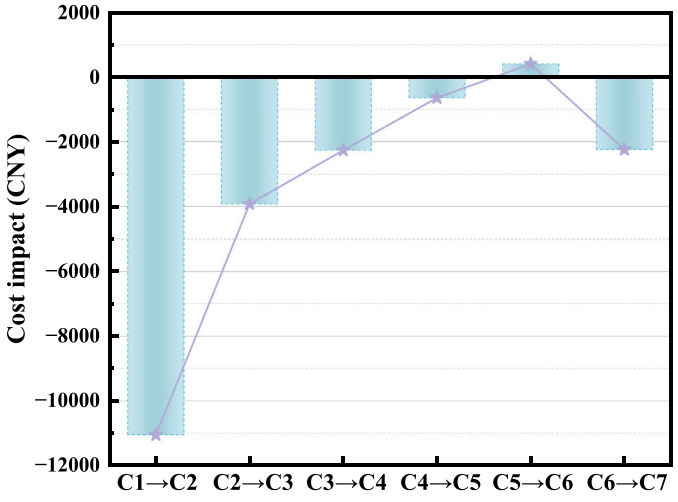


Fig. 5. Marginal cost impacts of model components.

7.3. SLA reliability under heterogeneous workload scheduling

Fig. 6 shows the SLA violation rate and SLA penalty of interactive workload and batch workload across the seven cases. From Case 1 to Case 7, the SLA violation rate of both workload types was reduced, and the total SLA penalty decreased by 75.4%. The lowest SLA violation rate and SLA penalty among the compared cases were obtained in Case 7, indicating that the proposed bi-timescale framework was associated with improved service reliability while coordinating data center workload scheduling with multi-microgrid operation.

The decomposition in Fig. 6 shows different SLA reliability characteristics for the two workload types. The batch workload had a higher SLA violation rate in all cases, whereas the SLA penalty was mainly contributed by the interactive workload. This result is consistent with the stricter service requirement of interactive workload and suggests the need to distinguish interactive workload from batch workload in the SLA constraint design. In particular, the transition from Case 2 to Case 3 produced a more pronounced reduction in the batch-workload SLA violation rate, which supports the role of temporal workload reshaping in absorbing delay-tolerant workload without directly relaxing the service requirement of interactive workload.

The case-wise trends further clarify the interaction between SLA reliability and power-computing-carbon coordination. After the carbon

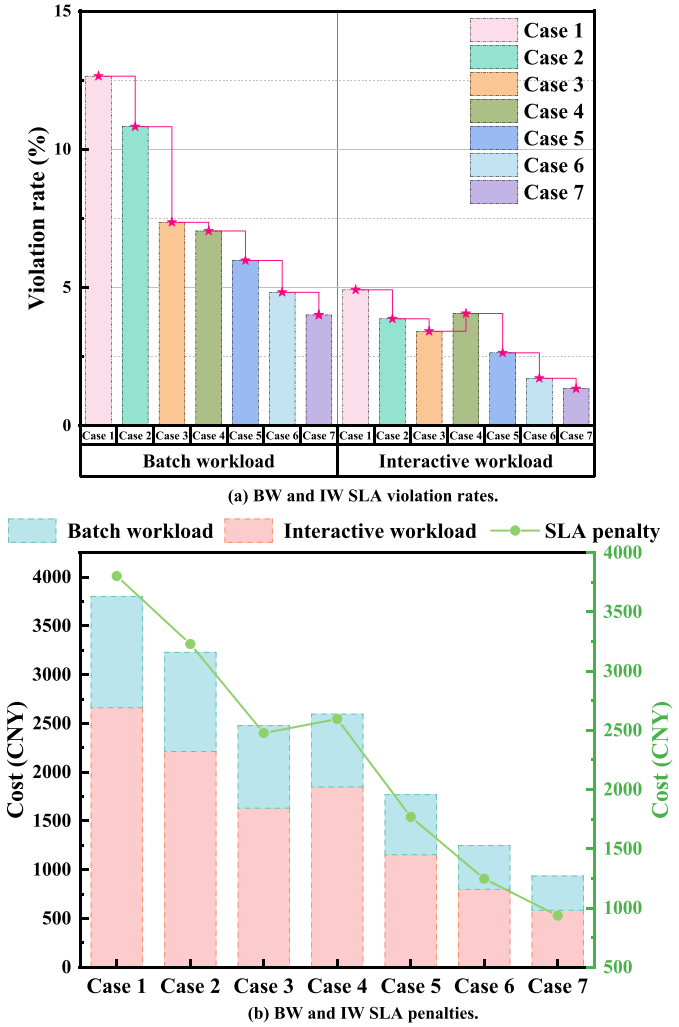


Fig. 6. SLA reliability and penalty decomposition across seven cases.

mechanism was introduced in Case 4, the total SLA penalty slightly increased relative to Case 3, mainly because the interactive-workload SLA penalty increased. This indicates that low-carbon dispatch alone was not sufficient to ensure SLA improvement. When scenario-based stochastic programming and the Wasserstein distributionally robust chance constraint were introduced in Cases 5 and 6, both workload types showed lower SLA violation rates and lower SLA penalties, indicating the value of uncertainty-aware scheduling for SLA reliability. With SAC-based server provisioning in Case 7, the SLA violation rates and SLA penalties were further reduced, suggesting that slow-timescale server provisioning can complement quasi-real-time robust scheduling in maintaining SLA performance.

7.4. Spatiotemporal flexibility in data center scheduling and multi-microgrid response

Fig. 7 shows the temporal reshaping of the total data center power profile under different flexibility settings. Compared with Case 2, where only spatial flexibility was enabled, Case 3 produced a flatter total data center power profile after temporal flexibility was introduced. The peak-to-valley range of the total data center power profile was substantially reduced, while the daily aggregate total data center power remained close to that of Case 2. This indicates that temporal workload reshaping mainly changed the timing of data center power demand rather than simply reducing the scheduled workload. Under the robust setting, Case 7 showed a lower aggregate total data center power than

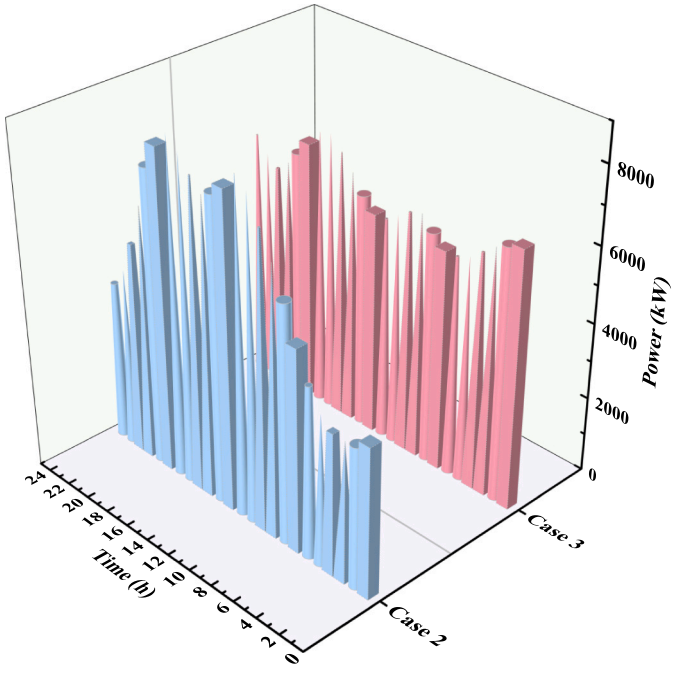


Fig. 7. Temporal reshaping of data center power.

Case 6, suggesting that SAC-based server provisioning can mitigate the power burden associated with fixed-server operation while retaining the quasi-real-time scheduling structure.

Fig. 8 further illustrates the spatial redistribution of equivalent data center power. In Case 1, the equivalent data center power was more concentrated in several data centers, whereas Case 2 exhibited a more balanced distribution across data centers after spatial workload migration was enabled. The daily share of equivalent data center power became less concentrated, especially with the reduction of the dominant contribution from data center 4 and higher shares assigned to other data centers. The distribution pattern of equivalent data-center power across locations is therefore reshaped under spatial flexibility, providing a spatial coordination basis for multi-microgrid scheduling.

Fig. 9 shows that spatiotemporal flexibility also affected the operation of the multi-microgrid system. From Case 1 to Case 2, spatial workload migration changed the balance between system grid purchase and local generation output. After temporal flexibility was further introduced in Case 3, the daily grid purchase was reduced relative to Case 2, while the local generation output increased. These results indicate that spatiotemporal flexibility was not confined to the data center side; rather, it reshaped the power demand seen by the microgrids and thereby influenced system-level operation. This coupling supports the interpretation that data center workload flexibility can serve as an operational resource for power-computing-carbon coordination in the proposed bi-timescale framework.

7.5. Carbon-aware coordination in multi-microgrid operation

Fig. 10 shows the carbon-aware dispatch response of Microgrid 4 by comparing Case 3 and Case 4. After the carbon mechanism was introduced, the local generator output of Microgrid 4 was generally reduced over the scheduling horizon, while grid purchase was adjusted accordingly. Since the local generation emission factor was higher than the grid emission factor in the plotted periods, the reduction in local-generation emission outweighed the increase in grid-purchase emission. As a result, the carbon emission of Microgrid 4 over the scheduling horizon was reduced in Case 4. This result indicates that the carbon mechanism

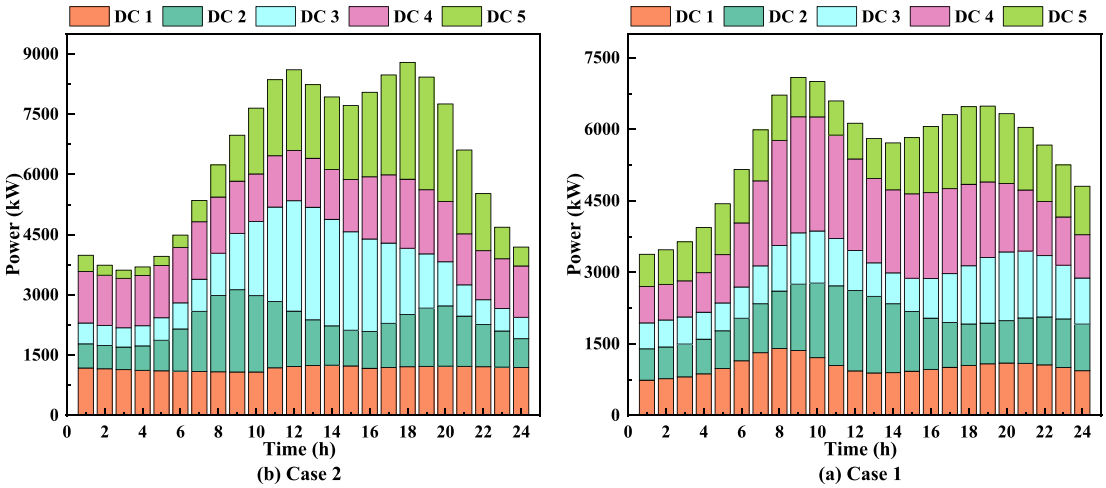


Fig. 8. Spatial redistribution of data center power.

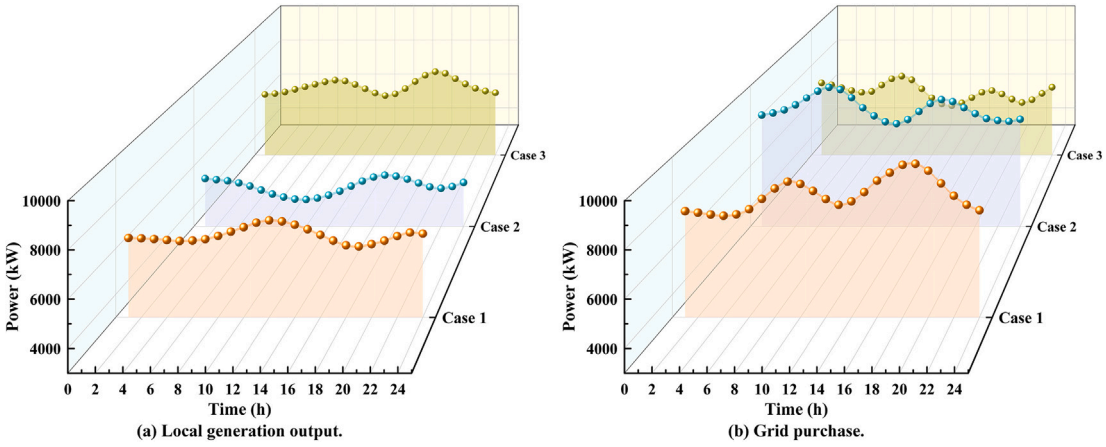


Fig. 9. Multi-microgrid operation response.

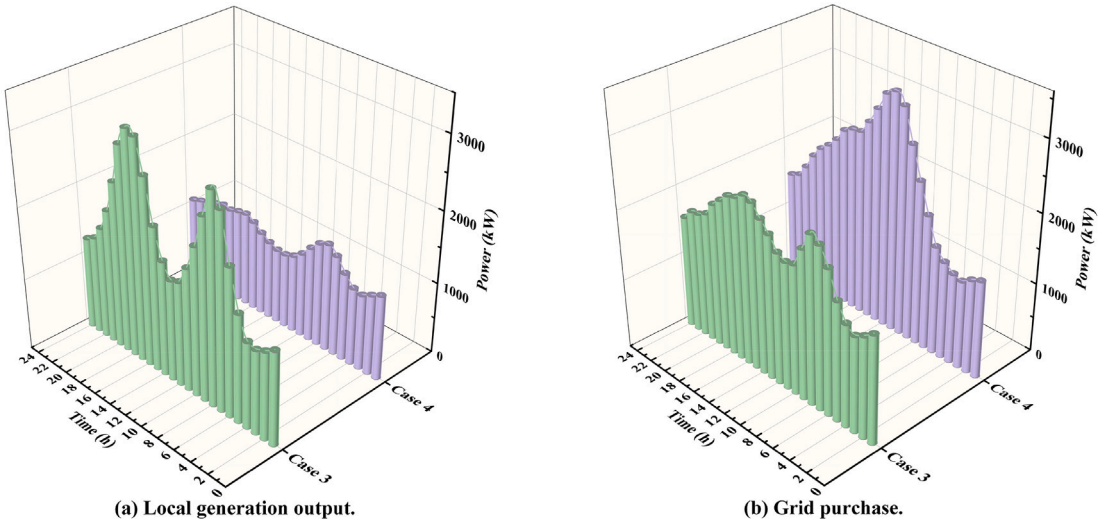


Fig. 10. Low-carbon dispatch response of Microgrid 4.

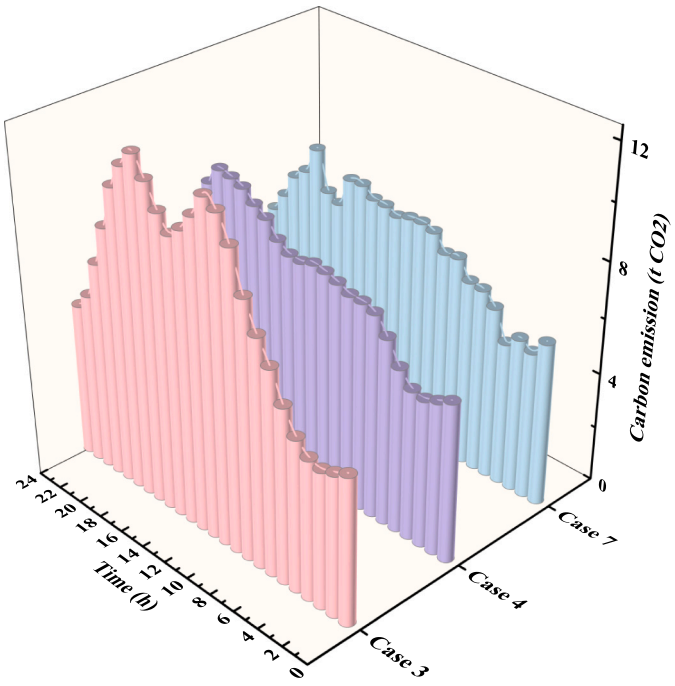


Fig. 11. System carbon emission across cases.

modified the microgrid dispatch pattern in a direction consistent with carbon-emission reduction.

At the system level, Fig. 11 shows the carbon emission under Cases 3, 4, and 7. Compared with Case 3, Case 4 achieved a reduction in system carbon emission over the scheduling horizon, supporting the effectiveness of carbon-aware scheduling at the multi-microgrid level. The complete coordinated framework in Case 7 further reduced the system carbon emission relative to Case 4, although the hourly carbon emission was not lower than that of Case 4 in every period. This observation indicates that the proposed power-computing-carbon coordination mainly improved the aggregate emission outcome rather than uniformly reducing carbon emission in each hour.

Taken together, these results indicate that carbon-emission reduction was obtained through both local dispatch adjustment and system-level coordination. The first comparison illustrates how the carbon mechanism changed the dispatch behavior of a representative microgrid, while the system-level comparison shows that the coordinated operation of microgrids and data centers reduced system carbon emission over the scheduling horizon. These results suggest that the carbon mechanism was not an isolated market add-on, but was integrated with data center workload scheduling and multi-microgrid operation within the proposed bi-timescale framework. At the same time, the non-uniform hourly pattern in Case 7 indicates that carbon reduction was achieved under operational tradeoffs with workload scheduling, SLA performance, and robust scheduling requirements.

7.6. Out-of-sample reliability and robustness-cost trade-off under wasserstein-DRCC

Fig. 12 compares the out-of-sample violation probability and daily operating cost under deterministic scheduling, scenario-based stochastic programming, and Wasserstein-DRCC. Compared with deterministic scheduling and scenario-based stochastic programming, Wasserstein-DRCC reduced the violation probabilities of both the power-balance constraint and the interactive-SLA constraint. The joint violation probability was also reduced to the lowest level among the compared methods. This indicates that Wasserstein-DRCC was associated with improved out-of-sample reliability for the coupled power-computing operation of the multi-microgrid system with data centers. Meanwhile, the daily operating cost under Wasserstein-DRCC was slightly higher than that under scenario-based stochastic programming, which reflects a limited robustness premium rather than a reliability improvement without cost impact.

Fig. 13 further illustrates the planned operating cost under different Wasserstein radius values. As the Wasserstein radius increased, the planned operating cost increased monotonically. This trend aligns with the interpretation that a larger ambiguity set necessitates a more conservative dispatch schedule. The selected Wasserstein radius was located in the middle of the tested range, where the cost increase was moderate compared to larger radius values. Consequently, the selected setting provides a practical balance between robustness and operating cost for the proposed power-computing-carbon coordination framework.

Fig. 14 shows the corresponding change in out-of-sample violation probability with respect to the Wasserstein radius. The violation probability decreased for the power-balance constraint, the interactive-SLA constraint, and their joint event as the Wasserstein radius increased. The selected Wasserstein radius provided a practical balance between reliability and operating cost, whereas further increases improved reliability only at a higher planned operating cost, as indicated in Fig. 13. These results support the role of Wasserstein-DRCC in controlling reliability risk under uncertainty, but they also show that the Wasserstein radius should be selected as a trade-off parameter rather than simply enlarged.

7.7. SAC-based server provisioning for adaptive idle-power control

Fig. 15 compares the system idle information technology power under fixed-server operation in Case 6 and SAC-based server provisioning in Case 7. Under the fixed-server setting, the idle information technology power remained almost unchanged over the scheduling horizon, which is consistent with a nearly fixed reserve computing capacity even when the workload-related power demand varied. In contrast, SAC-based server provisioning reduced the idle information technology power in every period and lowered the daily aggregate idle information technology power by about 13%. This result indicates that the SAC layer adjusted the server provisioning level according to time-varying operating conditions, rather than maintaining a nearly constant idle information technology power baseline.

The reduction in idle information technology power provides the computing-side explanation for part of the performance improvement from Case 6 to Case 7. Since idle server power directly contributes to data center power consumption, reducing unnecessary idle capacity can reduce the power burden imposed by data centers on microgrids. Within the proposed bi-timescale framework, SAC-based server provisioning was performed at the slow timescale, while the quasi-real-time scheduling layer retained Wasserstein-DRCC-based feasibility control. Therefore, the result in Fig. 15 supports the role of SAC in complementing robust scheduling: server capacity was adaptively provisioned to reduce idle information technology power, while the quasi-real-time scheduling layer was preserved for power-computing coordination.

7.8. Shapley-value allocation of multi-microgrid coalition cost savings

Table 2 reports the cost-saving allocation of the cooperative coalition under Case 7. The total stand-alone cost of the five microgrids is reduced from 44,597.2 CNY to 41,380.6 CNY over the scheduling horizon under cooperative operation, yielding a coalition saving of 3216.6 CNY. The Shapley-allocated savings sum to the total coalition saving, which indicates budget balance in the cooperative settlement.

The Shapley-value results show that the coalition saving was not assigned uniformly across microgrids. Larger allocated savings were associated with microgrids that provided stronger support for coordinated operation, such as low-carbon dispatch support, data center workload buffering, or high-load cost relief. In contrast, microgrids with

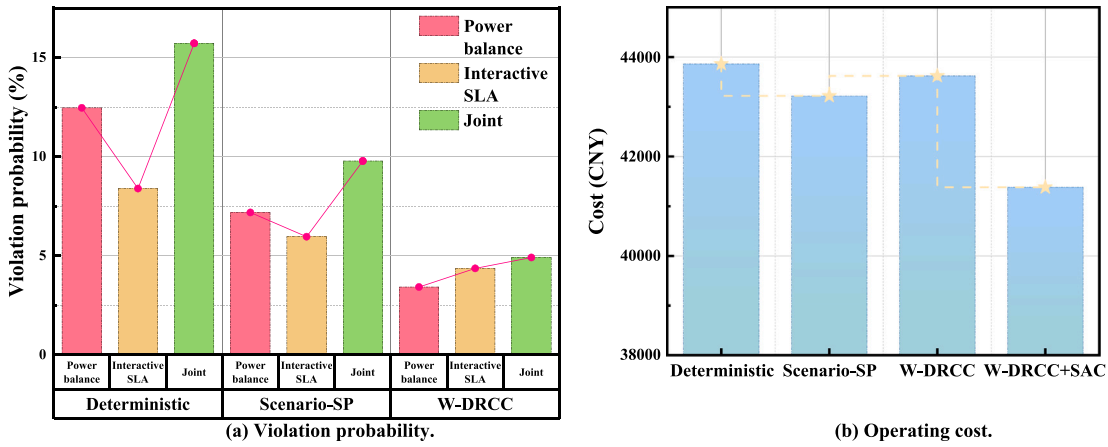


Fig. 12. Out-of-sample violation probability and operating cost.

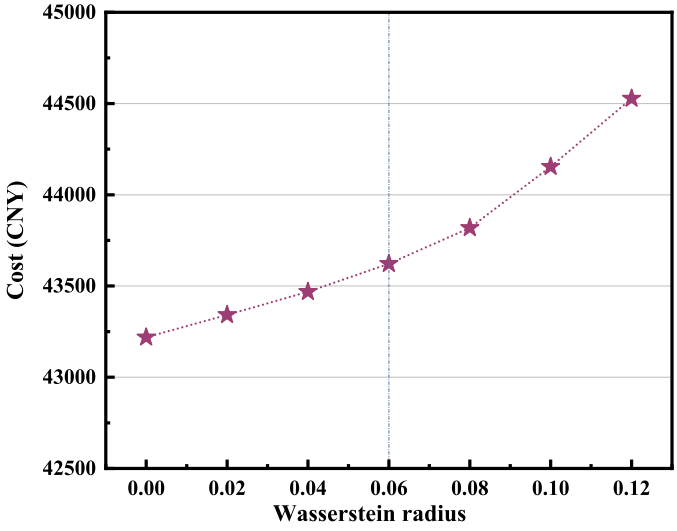


Fig. 13. Planned operating cost versus Wasserstein radius.

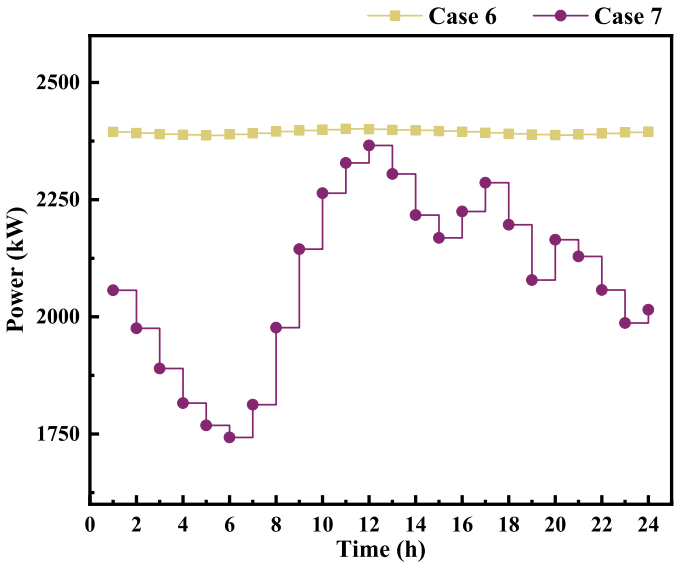


Fig. 15. Idle information technology power.

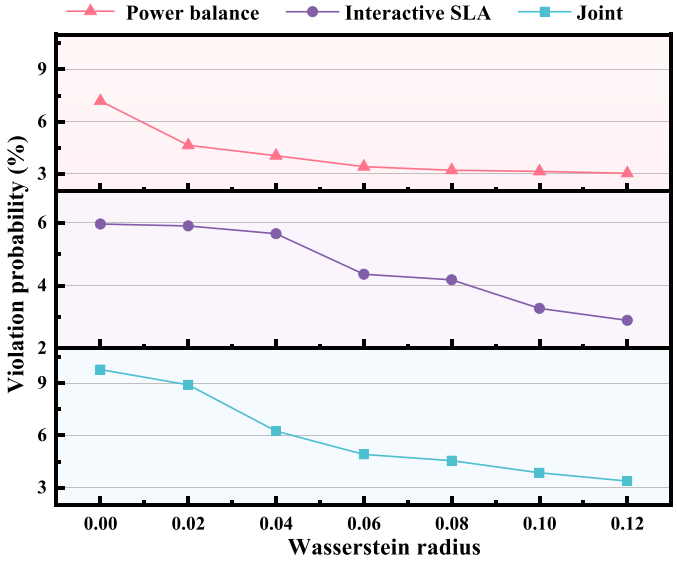


Fig. 14. Violation probability versus Wasserstein radius.

Table 2
Coalition cost saving and Shapley allocation.

Microgrid	Stand-alone cost (CNY)	Cooperative operating cost (CNY)	Coalition saving (CNY)
MG1	4212.37	4083.72	128.65
MG2	7588.95	6695.68	893.27
MG3	10,376.30	9289.36	1086.94
MG4	13,481.71	12,550.15	931.56
MG5	8937.88	8761.67	176.21
Total	44,597.2	41,380.6	3216.6

more limited balancing or migration capability received smaller but still positive allocated savings. All microgrids satisfy individual rationality, since each microgrid obtains a lower cooperative operating cost than its stand-alone cost.

These results support the use of the Shapley value as a contribution-aware cost-saving allocation mechanism for the proposed multi-microgrid coordination framework. The cost reduction generated by power-computing-carbon coordination can therefore be redistributed among microgrids in a way that preserves the total coalition saving and provides participation incentives for all members.

8. Conclusion

A bi-timescale distributionally robust framework was developed for power-computing-carbon coordination in multi-microgrid systems with data centers. The framework jointly coordinates spatial workload migration, temporal batch-workload deferral, cross-microgrid energy sharing, SLA requirements, carbon emission rights, and TGCs. SAC was used to determine day-ahead server counts, which were fixed in quasi-real-time scheduling to remove integer-continuous coupling and preserve the convex structure required by Wasserstein-DRCC. Multisource uncertainty from renewable generation, local load, and workload arrivals was captured through Wasserstein-DRCC with constraint screening, and the decomposed problem was solved by proximal adaptive ADMM. The resulting dual prices were further used for risk-aware settlement.

Case studies on a system with five microgrids and five data centers verified the effectiveness of the proposed framework. Compared with the baseline case, daily operating cost, carbon emission, and SLA penalties were reduced by 32.3%, 27.7%, and 75.4%, respectively. The results show that spatial workload migration provides major cost-saving potential, while temporal deferral further reshapes data center power demand and improves coordination with microgrid dispatch. The carbon mechanism also reduced aggregate emission when integrated with workload scheduling and energy sharing. Compared with deterministic and scenario-based stochastic scheduling, Wasserstein-DRCC improved out-of-sample reliability, with a limited robustness premium. SAC-based server provisioning also reduced daily aggregate idle information technology power by about 13%.

The settlement results indicate that cooperative operation can provide consistent incentives for participating microgrids. The total standalone cost was reduced from 44,597.2 CNY to 41,380.6 CNY, yielding a coalition saving of 3216.6 CNY. The sampled Shapley settlement maintained budget balance and individual rationality, while assigning larger savings to microgrids with stronger contributions to low-carbon dispatch, workload buffering, or high-load cost relief. These findings confirm that data center workload flexibility can be transformed into a schedulable resource for improving economy, carbon reduction, service reliability, and benefit allocation in multi-microgrid operation.

Several limitations remain. Network bandwidth and migration latency were treated as known parameters, although variable network conditions may affect feasible workload scheduling in practical data center operations. This issue can be addressed by incorporating uncertain network quantities into the uncertainty vector, motivating future work on robust scheduling under dynamic communication conditions. In addition, Shapley computation grows with coalition size. Although sampled permutations improve tractability, scalable settlement with accuracy-driven permutation selection should be further investigated. Future research should also develop adaptive Wasserstein-radius selection for non-stationary uncertainty and extend the framework to larger multi-microgrid systems with more detailed communication constraints and carbon-market mechanisms.

CRediT authorship contribution statement

Chenxian Song: Writing – review & editing, Writing – original draft, Supervision, Resources, Conceptualization. **Hui Chen:** Project administration. **Jian Wang:** Resources. **Yichen Sun:** Investigation. **Chunyang Gong:** Formal analysis. **Dikshit Chauhan:** Conceptualization. **Chuanlin Zhang:** Software. **Dipti Srinivasan:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported in part by the Key Program of the National Natural Science Foundation of China under Grant 62233006, and in part by the Youth Science Fund Project of the National Natural Science Foundation of China under Grant 52407124, and in part by Science and Technology Planning Project of Shanghai Science and Technology Commission under Grant 24DZ3001500.

Data availability

Data will be made available on request.

References

[1] International Energy Agency. Energy and AI. Paris: International Energy Agency; 2025 https://www.iea.org/reports/energy-and-ai [accessed 2 July 2026].
[2] Mytton D, Ashtine M. Sources of data center energy estimates: a comprehensive review. Joule 2022;6(9):2032–56.
[3] Nawaz A, Zhou M, Wu J, Long C. A comprehensive review on energy management, demand response, and coordination schemes utilization in multi-microgrids network. Appl Energy 2022;323:119596.
[4] Liu Z, Wang L, Ma L. A transactive energy framework for coordinated energy management of networked microgrids with distributionally robust optimization. IEEE Trans Power Syst 2020;35(1):395–404.
[5] Liu W, Yan Y, Sun Y, Mao H, Cheng M, Wang P, et al. Online job scheduling scheme for low-carbon data center operation: an information and energy nexus perspective. Appl Energy 2023;338:120918.
[6] Wang J, Deng H, Liu Y, Guo Z, Wang Y. Coordinated optimal scheduling of integrated energy system for data center based on computing load shifting. Energy 2023;267:126585.
[7] Poolla BK, Hota AR, Bolognani S, Callaway DS, Cherukuri A. Wasserstein distributionally robust look-ahead economic dispatch. IEEE Trans Power Syst 2021;36(3):2010–22.
[8] Arrigo A, Ordoudis C, Kazempour J, De Grève Z, Toubeau J-F, Vallée F. Wasserstein distributionally robust chance-constrained optimization for energy and reserve dispatch: an exact and physically-bounded formulation. Eur J Oper Res 2022;296(1):304–22.
[9] Zhou A, Yang M, Wang M, Zhang Y. A linear programming approximation of distributionally robust chance-constrained dispatch with Wasserstein distance. IEEE Trans Power Syst 2020;35(5):3366–77.
[10] Liu D, Luo Z, Qin J, Wang H, Wang G, Li Z, et al. Low-carbon dispatch of multi-district integrated energy systems considering carbon emission trading and green certificate trading. Renew Energy 2023;218:119312.
[11] Li Y, Zhou Y, Wang J, Ding J, Wei C, Liu Z. Co-optimization of configuration and energy scheduling of multi-communities microgrid systems considering carbon-green certificate trading mechanism. Energy 2026;344:139864.
[12] Boyd S, Parikh N, Chu E, Peleato B, Eckstein J. Distributed optimization and statistical learning via the alternating direction method of multipliers. Found Trends Mach Learn 2011;3(1):1–122.
[13] Mansouri SA, Nematbakhsh E, Ramos A, Tostado-Véliz M, Aguado JA, Aghaei J. A robust ADMM-enabled optimization framework for decentralized coordination of microgrids. IEEE Trans Ind Inform 2025;21(2):1479–88.
[14] Alonso Pedrero R, Pisciella P, Crespo del Granado P. Fair investment strategies in large energy communities: a scalable Shapley value approach. Energy 2024;295:131033.
[15] Alizadeh Bidgoli M, Ahmadian A. Multi-stage optimal scheduling of multi-microgrids using deep-learning artificial neural network and cooperative game approach. Energy 2022;239:122036.
[16] Han O, Ding T, Zhang X, Mu C, He X, Zhang H, et al. A shared energy storage business model for data center clusters considering renewable energy uncertainties. Renew Energy 2023;202:1273–90.
[17] Lian Y, Li Y, Zhao Y, Yu C, Zhao T, Wu L. Robust multi-objective optimization for islanded data center microgrid operations. Appl Energy 2023;330:120344.
[18] Zeng B, Zhou Y, Xu X, Cai D. Bi-level planning approach for incorporating the demand-side flexibility of cloud data centers under electricity-carbon markets. Appl Energy 2024;357:122406.
[19] Zheng X, Chen H, Xu Y, Li Z, Lin Z, Liang Z. A mixed-integer SDP solution to distributionally robust unit commitment with second order moment constraints. CSEE J Power Energy Syst 2020;6(2):374–83.
[20] Ding Y, Morstyn T, McCulloch MD. Distributionally robust joint chance-constrained optimization for networked microgrids considering contingencies and renewable uncertainty. IEEE Trans Smart Grid 2022;13(3):2467–78.
[21] Mu C, Shi Y, Xu N, Wang X, Tang Z, Jia H, et al. Multi-objective interval optimization dispatch of microgrid via deep reinforcement learning. IEEE Trans Smart Grid 2024;15(3):2957–70.
[22] Sun W, Zou Y, Zhang X, Guo N, Zhang B, Du G. High robustness energy management strategy of hybrid electric vehicle based on improved soft actor-critic deep reinforcement learning. Energy 2022;258:124806.
[23] Wang W, Wang B, Liu Z, Zheng J, Ma H, Li T, et al. A Stackelberg-generalized Nash equilibrium model for energy trading in distribution networks with novel two stage Wasserstein distributionally robust reformulation. Appl Energy 2026;413:127683.

[24] Shen Y, Zhai J, Kang Z, Zhao B, Gao X, Li Z. Distributionally robust chance-constrained energy management for island DC microgrid with offshore wind power hydrogen production. Energy 2025;316:134570.
[25] Esfahani M, Alizadeh A, Amjady N, Kamwa I. A distributed VPP-integrated co-optimization framework for energy scheduling, frequency regulation, and voltage support using data-driven distributionally robust optimization with Wasserstein metric. Appl Energy 2024;361:122883.
[26] Jin X, Liu B, Liao S, Cheng C, Zhang Y, Zhao Z, et al. Wasserstein metric-based two-stage distributionally robust optimization model for optimal daily peak shaving dispatch of cascade hydroplants under renewable energy uncertainties. Energy 2022;260:125107.
[27] Tang Y, Ding S, Lu Z, Lu S, Zhang J, Cai Y, et al. Multi-objective hybrid game-theoretic framework with distributionally robust chance constraints for multi-energy microgrid energy management. Appl Energy 2026;404:127184.
[28] Xu D, Liu Z, Hou K, Wei X, Jia H, Wu Z, et al. Distributionally robust frequency constrained low-carbon scheduling considering flexibility of carbon capture power plants. Energy 2025;332:137171.
[29] Zhang Y, Wu Q, Ren H, Li Q, Zhou W. Optimal operation of multi-microgrid systems considering multi-level energy-certificate-carbon coupling trading. Renew Energy 2024;227:120505.
[30] Yan M, Shahidehpour M, Alabdulwahab A, Abusorrah A, Gurung N, Zheng H, et al. Blockchain for transacting energy and carbon allowance in networked microgrids. IEEE Trans Smart Grid 2021;12(6):4702–14.
[31] Ju L, Lv S, Zhang Z, Li G, Gan W, Fang J. Data-driven two-stage robust optimization dispatching model and benefit allocation strategy for a novel virtual power plant considering carbon-green certificate equivalence conversion mechanism. Appl Energy 2024;362:122974.
[32] Xu J, Yi Y. Multi-microgrid low-carbon economy operation strategy considering both source and load uncertainty: a Nash bargaining approach. Energy 2023;263:125712.
[33] Fan W, Fan Y, Liu P, Wang Y, Tong F, Yi B, et al. Distributionally robust optimization scheduling model for electric power and computing power coordination considering spatiotemporal response. Appl Energy 2025;402:126895.
[34] Wang Y, Lin J, Han Y, Han K, Han J, Han T, et al. Comprehensive evaluation of all-element flexibility resources in data centers: considering synergistic benefits of computing, electricity, and heat. Appl Energy 2025;399:126442.
[35] Amaris H, Molina YP, Alonso M, Luyo JE. Loss allocation in distribution networks based on Aumann–Shapley. IEEE Trans Power Syst 2018;33(6):6655–66.
[36] Zhao D, Wang H, Huang J, Lin X. Virtual energy storage sharing and capacity allocation. IEEE Trans Smart Grid 2020;11(2):1112–23.